Prostate Cancer

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 153	Epidemiology, Aetiology & Prevention
Ch 154	Prostate Cancer Biomarkers
Ch 155	Prostate Biopsy
Ch 156	Pathology of Prostatic Neoplasia
Ch 157	Diagnosis & Staging
Ch 158	Active Management of Localised Disease
Ch 159	Active Surveillance
Ch 160	Open Radical Prostatectomy
Ch 161	Laparoscopic & Robotic Prostatectomy, PLND
Ch 162	Radiation Therapy
Ch 163	Focal Therapy
Ch 164	Locally Advanced Prostate Cancer
Ch 165	Recurrent & Newly Metastatic Disease
Ch 166	Castration-Resistant Prostate Cancer
★	Putting it together

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

Why this matters: who gets and who dies of the disease decides whom we screen.

- It is the **second commonest** non-skin cancer and the **fifth** cause of cancer death in men; 2020 saw ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest). US lifetime risk is ≈ **1 in 8**. 5-year survival is 97%, yet it is the second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence follows screening and wealth. Mortality peaks where African ancestry meets poor access: the **Caribbean has the highest mortality in the world** MCQ. Asia is lowest on both but rising.
- **Latent (autopsy) cancer** is ~5% under 30 y, 29% at 60–69 and **59% >79 y**. It is higher in Black men at every age.

> **KEY POINT**
> Aetiology is hard to study: in a prostate-specific antigen (PSA)-screened population a diagnosis mostly means a man "happened to have a PSA test", so risk-factor studies may find determinants of *being tested*. The preferred end point is **aggressive disease** (advanced, metastatic, death). **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits falsely seem to raise incidence while lowering mortality.

2 · Established risk factors

Why this matters: these factors decide who starts screening earlier.

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean age at diagnosis is 66; ~60% are ≥65; it is rare under 40. Diagnosis at ≥70 carries a higher risk of cancer death.
- **Race**: Black men have the highest incidence (172.6/100,000 in the US) and **twice** the cancer-specific mortality of White men. They present younger and more metastatic.
- Drivers: unequal access, comorbidity and **8q24** (and 17q21) alleles. With **equal access, outcome is equal** in non-metastatic disease.
- **Family history** is present in ~20%. Twin-study heritability is up to **58%**, so genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative (father or brother)	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

Why this matters: an inherited repair defect changes screening, surveillance and drugs.

The key inherited genes are **DNA-repair genes**, which predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; poly(ADP-ribose) polymerase (PARP) inhibitor; caution in active surveillance (AS)
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very sensitive to androgen deprivation therapy (ADT)
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	Genome-wide association study (GWAS) loci	Commoner in African ancestry	Explain part of the racial excess

- **Polygenic risk scores** predict *any* cancer, not aggressive cancer, so they are not yet practice-changing.
- P4 Indications of genomic testing in carcinoma prostate — *germline testing: metastatic, high-risk, intraductal/cribriform, family history, Ashkenazi*

Not in the paper answer:

- GWAS has found >200 single-nucleotide polymorphisms (SNPs), each of small effect.
- Steroid-pathway SNPs (HSD3B1, CYP19A1, HSD17B4) link with progression; KLK3 (PSA gene) SNPs with risk.

4 · Molecular aetiology and inflammation

Why this matters: this explains how cancer starts and why populations differ.

- **Chronic inflammation** is in nearly every biopsy; it raises PSA, but evidence supports a causal role.
- † **Proliferative inflammatory atrophy (PIA)** is the proposed precursor, progressing via high-grade prostatic intraepithelial neoplasia (HGPIN) to cancer.
- **TMPRSS2-ERG** fusion puts the ERG oncogene (ETS family) under an androgen-regulated promoter. It is in ~half of Western tumours but uncommon in East Asians; FOXA1 mutations are commoner in Chinese men. *See Ch 154*

5 · Possible risk factors — body, metabolism, lifestyle

Why this matters: patients ask what they can change; the answer concerns aggressive disease.

Most lifestyle factors act on **progression** rather than initiation; evidence is observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% higher per 10 cm	More high-grade cancer and death
Obesity	Mixed; less low-grade	**Consistently raised**

Factor	Total incidence	Aggressive / fatal
Smoking	Not raised (smokers screen less)	**Mortality raised ~40%**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, cardiovascular disease (CVD)	Slightly raised	Metabolic syndrome: worse pathology
Diabetes	Lower (low testosterone, IGF-1 and PSA; metformin)	—
Vigorous activity	No clear effect	Less advanced disease
Alcohol	Unclear	Mortality raised at ≥45 g/day; binge drinking raises it

- **Mechanism**: adiposity raises insulin, insulin-like growth factor-1 (IGF-1), leptin, inflammation and oestrogen, and lowers sex hormone-binding globulin (SHBG). This drives proliferation and blocks apoptosis.
- **Adult serum testosterone does *not* predict risk** MCQ. Genetically higher free testosterone, IGF-1 and early puberty modestly do.
- A **composite healthy lifestyle** means **30–50%** less lethal cancer. In the highest polygenic-risk quartile, lifetime lethal risk falls from 5.3% to 1.6%.
- On ADT, do a baseline cardiovascular assessment (the CVD link is observational only).
- **Diet**: World Cancer Research Fund (WCRF) 2018 found **no dietary factor with strong evidence**. PSA-era downgrades: dairy and calcium to "limited" and lycopene to "no conclusion". Lycopene needs tomato cooked in oil.
- **Vasectomy** is not a risk factor (detection bias); the American Urological Association (AUA) says it need not be discussed.
- Transgender women on oestrogen have a median PSA ≈ **0.02 ng/mL** (50-fold lower), so cut-points must be lowered.

6 · Supplements

Why this matters: many men take supplements, and some harm.

- P4 Antioxidants in urological malignancy; Prostate cancer prevention trials — *SELECT null, vitamin E harmful; VITAL null; lycopene, green tea, soy/pomegranate/curcumin*

Not in the paper answer:

- Calcium supplements raise fatal cancer (relevant on ADT).
- Zinc ≥75 mg/day or for ≥15 y raises lethal cancer.
- Pomi-T (pomegranate, green tea, turmeric, broccoli) slowed PSA rise in a randomised trial.
- No supplement benefits healthy men on a balanced diet.

> **KEY POINT**
> Vitamin E worked in the Alpha-Tocopherol, Beta-Carotene (ATBC) trial but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

Why this matters: many men already take these drugs for benign enlargement.

- P4 Prostate cancer prevention trials — *5ARI rationale, PCPT/REDUCE high-grade controversy, FDA warning, no mortality effect; aspirin, statins, metformin*

Not in the paper answer:

- Real-world data (>3 million men): fewer diagnoses, more Gleason 8–10, no mortality effect.
- Statins have never been tested in a prevention randomised trial.
- No preventive agent is approved by the Food and Drug Administration (FDA); the bar is now preventing **aggressive** cancer. 5α-reductase inhibitors (5ARIs) shrink the gland about 25%, and on a 5ARI you **double the PSA**. *See Ch 154*
- **Secondary prevention**: in the Prostate Cancer Lifestyle Trial (men on AS), plant-based diet, exercise and stress management meant **5% vs 27%** treated at 2 y.
- Activity lowers cancer-specific mortality ~31% (≥150 min/week moderate or ≥75 min vigorous). A Western diet raises cancer death (hazard ratio 2.5), high-fat dairy raises recurrence, and continued smoking raises recurrence and death.

KEY POINT

Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does, so lifestyle advice benefits both.

8 · PSA screening

Why this matters: screening saves lives but causes overdiagnosis.

- P2 PSA screening for prostate cancer: evidence, problems and solutions — *trials, corrected ERSPC benefit, overdiagnosis, USPSTF D to C, MRI-first*
- Screening lowers cancer death ~**20–30%**, and the benefit builds with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC (European Randomized Study of Screening for Prostate Cancer)	>182,000; PSA every 2–4 y vs none	Cancer death fell 16% (9 y), then ~20% (16 y); excess incidence fell 85% to 36%	Benefit is time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes the effect
Göteborg-1	Swedish screening	Significantly fewer cancer deaths	Supports ERSPC; basis of 4Kscore
† **CAP** (UK)	Single PSA invitation	No benefit at 10 y; tiny fall in death at 15 y	One-off screening is weak
PCPT (Prostate Cancer Prevention Trial)	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer fell **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death fell 25% (not significant)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer fell **22.8%**; Gleason 8–10 0.9 vs 0.6% (years 3–4: 12 vs 1)	FDA warning
SELECT	35,533; selenium 200 µg with or without vitamin E 400 IU	Stopped for futility; vitamin E raised risk (P = 0.06)	No supplements

Trial	Design	Result	Take-home
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

Why this matters: PSA alone sends too many men to biopsy.

- Prostate-specific antigen (PSA) is **organ-specific, not cancer-specific**, which causes negative biopsies and overdiagnosis of Grade Group (GG) 1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; the target is **clinically significant cancer (GG ≥2)**. *See Ch 153*
- Settings: before initial or repeat biopsy; favourable-risk cancer (active surveillance (AS) vs treatment); after radical prostatectomy (RP), for adjuvant or salvage decisions.
- Generations: organ (PSA), then cancer (PCA3, TMPRSS2:ERG, kallikreins), then high-grade cancer (SChLAP1).

2 · PSA biology

Why this matters: it explains every derivative and confounder.

- PSA is **hK3** (gene KLK3 on **19q13**), an androgen-regulated serine protease that **liquefies semen** by cleaving semenogelin. Half-life is **2–3 days**.
- It is made as **proPSA** (244 amino acids (aa), 7-aa leader); **hK2 cleaves the leader** to give active PSA (237 aa).
- Serum PSA rises when it leaks through a lost basal layer and basement membrane. Per cell, cancer makes *less* PSA than benign tissue, but pattern 4 adds more serum PSA per cm^3.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No**: epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, benign PSA (BPSA), intact PSA)

Confounders

- After biopsy the rise lasts ≥**4 weeks**; cycling >55 km adds ~10%; ejaculation raises it for 24 h, so repeat after abstinence.
- **5α-reductase inhibitors (5ARIs)** lower PSA ~**50%** by 12 months, so **double** it. This overestimates in the first 6 months and underestimates after years.
- † Age-specific upper limits (ng/mL): 40–49, 2.5; 50–59, 3.5; 60–69, 4.5; 70–79, 6.5.

3 · PSA derivatives

Why this matters: cheap refinements of PSA can spare a biopsy.

Derivative	Definition / values	Use
PSA density	PSA ÷ volume; cut-off **0.15** (0.08–0.15)	After negative/equivocal magnetic resonance imaging (MRI); ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year	Minimal value; **never an isolated biopsy trigger**

Derivative	Definition / values	Use
% free PSA	Cancer PSA binds ACT, so **low %fPSA suggests cancer**; thresholds 15–25%	Food and Drug Administration (FDA): PSA 4–10, benign digital rectal examination (DRE); mainly **after negative biopsy**; unaffected by 5ARI

KEY POINT
%fPSA was validated against *any* cancer, not GG ≥2, so it overestimates significant risk. Risk calculators (PCPT, Prostate Biopsy Collaborative Group) include it.

- **[-2]proPSA** comes from the peripheral zone and rises with Gleason, so it underpins the Prostate Health Index (PHI). **BPSA** comes from the transition zone and tracks benign hyperplasia, so it has no prognostic value. **Intact PSA** is in the 4Kscore.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Why this matters: these National Comprehensive Cancer Network (NCCN) tests avoid needless biopsies.

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA and **hK2**, with age, DRE, prior biopsy	**7.5%**: sensitivity 94%, negative predictive value (NPV) 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 messenger RNA (mRNA) with clinical data	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3, TMPRSS2:ERG and PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3** is a long non-coding RNA (9q21–22). Score = (PCA3 ÷ KLK3 mRNA) × 1000 after an **attentive DRE**; thresholds 25/35. It is **independent of prostate size** and FDA-approved after a **previous negative biopsy**.
- **TMPRSS2:ERG** is **~100% specific** but in only ~half of tumours, so it is combined with PCA3.
- hK2 rises with cancer and grade independently of PSA.

KEY POINT
Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra. ExoDx avoids this, as exosomes are shed spontaneously. Urinary markers are highly specific; their weakness is coverage, hence panels.

5 · Tissue-based biomarkers

Why this matters: tissue tests guide re-biopsy, surveillance and post-RP radiotherapy.

- P2 Molecular staging of CA prostate — *Decipher, Oncotype GPS, Prolaris, PORTOS, ConfirmMDx; limits: heterogeneity, retrospective, G-MAJOR*

Not in the paper answer:

- Prolaris uses 31 cell-cycle plus 15 housekeeping genes and also predicts recurrence after RP or radiotherapy.
- Prolaris and Oncotype are used on biopsies (AS vs treatment); Decipher, built for post-RP use, is now also used on biopsies.
- **ConfirmMDx** measures GSTP1, APC and RASSF1 methylation in negative cores (NPV ~88–90%), using the **field ("halo") effect** around a missed tumour.
- Oncotype predicts primary pattern 4 or pT3 at RP. **Decipher ≥0.4 post-RP** suggests benefit from radiotherapy.

6 · Germline testing

Why this matters: inherited mutations change screening for the man and his family.

- Up to 40–50% of risk is familial, mostly via **DNA-damage-repair** genes. STHLM3 combines 4 kallikreins, 232 single-nucleotide polymorphisms and clinical variables.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 odds ratio ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract
HOXB13 G84E	4.5×	**Not** more aggressive

Next-generation sequencing tests BRCA1/2 alone or a panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications: *see Ch 153*.

7 · Clinical algorithm

ALGORITHM

1. **Raised PSA**: repeat it, excluding infection, instrumentation, ejaculation and 5ARI. Then do MRI and/or a biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density). If risk is high, biopsy.
2. **Negative biopsy, persistent suspicion**: MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx. If positive, targeted re-biopsy.
3. **Favourable-risk**: PSA density and Prolaris/Oncotype/Decipher decide AS vs treatment.
4. **After RP**: † PSA should be undetectable. If ≥0.2 ng/mL and confirmed, this is biochemical recurrence; then Decipher/PORTOS guides adjuvant or salvage radiotherapy.

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

Why this matters: imaging cannot diagnose cancer; only tissue can.

On transrectal ultrasound (TRUS) the peripheral (PZ) and central zones are homogeneous and echogenic; the transition zone (TZ) is heterogeneous. Most cancers arise in the posterolateral PZ. **39%** of cancers are isoechoic, so a normal grey scale never precludes biopsy.

- P2 TRUS / prostate biopsy — *indications, contraindications, preparation, technique, TR vs TP, complication rates*

Not in the paper answer:

- Stop herbal supplements; biopsy at international normalised ratio (INR) <1.5.
- Admitted sepsis: carbapenem for up to 2 weeks.
- **Initial** biopsy also follows cancer found at transurethral resection. **Repeat** biopsy: rising prostate-specific antigen (PSA); atypical small acinar proliferation (ASAP; **≈50%** cancer risk); atypical intraductal proliferation; positive PCA3 or ConfirmMDx; suspicious MRI; recurrence after ablation; active surveillance (AS).
- National Comprehensive Cancer Network (NCCN): biopsy for abnormal digital rectal examination (DRE) with raised PSA, or repeat PSA >3 ng/mL with Prostate Imaging Reporting and Data System (PI-RADS) ≥3; PSA velocity alone, or abnormal DRE with normal PSA, is not enough. *See Ch 154*
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × anteroposterior × length (1 cm^3 ≈ 1 g).
- **Contraindications**: coagulopathy, severe immunosuppression, acute urinary tract infection (UTI) or prostatitis. Painful anorectal disease needs anaesthesia; with no anus, use in-bore MRI or transperineal biopsy.

2 · Pre-biopsy mpMRI

Why this matters: MRI first finds more significant cancer and spares biopsies.

- P1 P4 Prostate multiparametric MRI; PI-RADS and MRI–ultrasound fusion biopsy — *sequences, PI-RADS scoring, zone rules, cancer rate per score, targeting methods*

Not in the paper answer:

- A third of MRI-missed cancers are primary pattern 4.

Multiparametric magnetic resonance imaging (mpMRI: T2-weighted (T2WI), diffusion-weighted (DWI) and dynamic contrast-enhanced (DCE)) is a **triage test**: with no lesion biopsy may be avoided, and targeting finds more high-grade, fewer low-grade cancers. A lesion **>1.5 cm** or definite extraprostatic extension (EPE) scores PI-RADS 5. American Urological Association/Society of Abdominal Radiology (AUA/SAR) 2016: mpMRI and targeting of PI-RADS ≥3 after a negative biopsy with rising PSA.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	Positive DCE upgrades to 4	DWI 5 upgrades to 4
† v2.1 (2019)	—	T2WI 2 with DWI ≥4 becomes 3

> **KEY POINT**
> DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4): MRI-invisible lesions, reader discordance, registration or needle error, small lesion in a large gland. So systematic cores remain relevant.

- Targeting is cognitive (position unrecorded), software fusion (UroNav electromagnetic; Artemis mechanical arm) or in-bore. Fusion-targeted cores are ≈3× likelier than systematic ones to find intermediate/high-risk disease; PI-RADS 5 targets are nearly always positive.
- Take **2–4 cores per target** (AUA/SUO 2023: 2–3, with or without systematic cores) whenever biopsy is indicated and a PI-RADS ≥3 lesion exists; also in AS and focal-therapy planning.

3 · Systematic templates and technique

Why this matters: where the needle goes beats how many cores.

- **Sextant** biopsy (Hodge 1989) sampled too much TZ and missed lateral PZ. **Extended 12-core** is the AUA standard. Going from 6 to 12 cores raises detection; 18–21 do not: **geometry, not core number**.
- Add TZ/anterior cores for repeat biopsy or gland >50 mL; seminal vesicle cores for a palpable abnormality.
- 18-gauge gun (throw 15–25 mm; tip runs 0.5 cm beyond the core): indent the capsule, withdraw a few mm, fire. Never reposition with the needle on the rectal wall. ≤2 cores per jar.

4 · Transrectal vs transperineal (TP)

Why this matters: the perineal route avoids rectal bacteria.

- TP passes clean perineal skin (transrectal is Class III), causes no rectal bleeding and reaches apex and anterior gland (anterior-only cancers 6–20%). Templates: Ginsburg; MUSIC 12-core, which omits midline cores so causes less pain and haematuria. None is universal.
- European Association of Urology 2023: TP **preferred** (strong); detection non-inferior ("TREXIT"). Give skin-flora cover or none; omitting antibiotics did not change the very low sepsis rate.
- Access: brachytherapy grid (5-mm mapping), freehand guide (PrecisionPoint) or two 14-gauge cannulas (cheapest; learning curve 4–6 weeks).

5 · Antibiotic prophylaxis

Why this matters: sepsis from resistant bowel organisms is the dangerous risk.

- Every transrectal patient gets prophylaxis. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin with or without aminoglycoside; or 3rd-generation cephalosporin. Give ≥1 h before, for **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Aztreonam replaces aminoglycoside in renal impairment.
- Food and Drug Administration (FDA) 2016 fluoroquinolone warning: tendon rupture, neuropathy, central nervous system effects, myasthenia.
- Infection risk factors: antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI or biopsy infection, many prior biopsies, travel (India, South-

East Asia), health-care work, rectal resistant *E. coli*. They need augmented or swab-targeted prophylaxis, or TP.

- Endocarditis risk: IV ampicillin (vancomycin if allergic) and gentamicin, then oral quinolone 2–3 days. Enema: no proven infection benefit.

6 · Anticoagulation and anaesthesia

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
Direct oral anticoagulants (DOACs)	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at the seminal vesicle–prostate junction (beats intrarectal gel).
- Transperineal: 1–2 mL to the skin, then 5 mL into the **periapical triangle** (levator ani lateral, rhabdosphincter medial, external anal sphincter posterior, apex superior), then 3 mL along the capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%): catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics
Sepsis	**0.3–3.1%**	Carbapenem until cultures
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

1. **Post-biopsy fever**: assess for sepsis and take cultures.
2. Give **no empirical fluoroquinolone**: *E. coli* is resistant (also to co-trimoxazole, ampicillin). Start a carbapenem (or amikacin/3rd-generation cephalosporin by sensitivity) and resuscitate.

ALGORITHM

1. **Persistent rectal bleeding**: apply pressure.
2. If it continues, tamponade (sheath or Foley balloon). If still bleeding: endoscopy with adrenaline/sclerosant, then angioembolisation, then transrectal suture.

KEY POINT

Commonest complication: bleeding overall; infection among serious ones and admissions (1.0% rising to 4.1% in the resistance era). Frequent-trivial vs rare-dangerous explains universal prophylaxis and the move to the perineum.

8 · Repeat and saturation biopsy

Why this matters: each negative round finds less cancer.

- P2 Saturation biopsy in CA prostate — *>12 cores, template mapping, indications, no gain at initial biopsy*

Not in the paper answer:

- Micro-ultrasound (29 MHz; PRI-MUS score) approaches mpMRI without contrast (emerging).

ALGORITHM

1. Cancer yield on successive biopsies in the European Randomized Study of Screening for Prostate Cancer (ERSPC): **22% → 10% → 5% → 4%**.

- Take a third or fourth biopsy only for high suspicion, using mpMRI with or without targeting, ConfirmMDx, anterior/TZ cores and the TP route. Transurethral biopsy is obsolete.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; negative predictive value 89% vs 74%; 27% avoid biopsy	mpMRI better triage
PRECISION (2018)	500 biopsy-naive: MRI with targeted biopsy vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
† MRI-FIRST (2019)	Paired systematic and targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)†	TP without antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest false-positive hypoechoic lesion: benign prostatic hyperplasia (BPH) foci and prostatitis.
2. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.
3. Absent seminal vesicle: ipsilateral renal agenesis in 79%.
4. Müllerian cysts are midline; wolffian cysts are lateral.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

Why this matters: some lesions need no action; others mean high-grade cancer is present.

- P2 PIN — *definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified*

Not in the paper answer:

- IDC-P is clonally related to the invasive cancer; rarely it is a high-grade precursor.
- ~**20%** of high-grade prostatic intraepithelial neoplasia (HGPIN) carries the **TMPRSS2:ERG** fusion. Isolated HGPIN (unifocal or multifocal, or on transurethral resection (TUR)) needs no repeat biopsy once multiparametric magnetic resonance imaging (mpMRI), digital rectal examination (DRE) and prostate-specific antigen (PSA) are weighed.
- **Intraductal carcinoma of the prostate (IDC-P)** shows atypia beyond HGPIN (dense cribriform) with **basal cells retained**. It is usually **retrograde spread of high-grade invasive cancer**. Prevalence rises from ~2% in low risk to >50% in metastatic disease. It independently predicts early biochemical recurrence (BCR). Grade Group (GG) 1 with IDC-P is upgraded at radical prostatectomy (RP) in **>80%**. Strict (Guo–Epstein) criteria permit definitive therapy without proven invasion.

Lesion	Defining feature	Action on isolated finding
Low-grade PIN (LGPIN)	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN with adjacent small atypical glands	Manage as atypical
Atypical intraductal proliferation (AIP)	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes active surveillance (AS)**

KEY POINT

HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present.
Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

Why this matters: tumour site explains what DRE and biopsy miss.

- **Zones (McNeal)**: †peripheral zone (PZ): ~70% of gland and of cancers, including the main mass in T2 and **85%** of T1c; posterolateral and palpable.
- †Transition zone (TZ): ~5% of gland, ~20% of cancers; origin of benign prostatic hyperplasia (BPH). Its tumours are anterior, impalpable, under-sampled and lower grade.
- †Central zone (CZ), around the ejaculatory ducts: ~25% of gland, ~5% of cancers, rare but aggressive (seminal vesicle invasion, SVI). Periurethral glands are <1% (median lobe).
- **Anterior tumours** are ~15% of RPs; their extraprostatic extension (EPE) is more often margin-positive. Tumours are **multifocal in >85%**.

- With no true capsule, say "**extraprostatic extension**". Tumour exits via **perineural spaces**, posterolaterally.
- **SVI** means tumour in the SV *muscular wall*. Usually tumour leaves at the base into periseminal tissue, then the SV; less often it tracks along the ejaculatory ducts.
- Metastases: nodes and bone; lung at autopsy.
- EPE is uncommon below **0.5 cm³**; nodes and SVI are uncommon below **4 cm³**.

Stage	Definition
T1a / T1b	Incidental at TUR of prostate (TURP)/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, found on needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at an intraprostatic incision or ill-defined boundary (apex)
pT3a	EPE; **includes microscopic bladder neck invasion**
pT3b	SVI
pT4	†Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck invasion is pT3a; T1a/b depend on the 5% cut-off, not grade.

3 · Grading: Gleason and grade groups

Why this matters: grade is the strongest biopsy predictor.

- P2 Gleason's score and its significance — *history, scoring rules, significance, limitations*

Not in the paper answer:

- Expanded pattern 4 moved many old 6s to 7, so modern 6 is uniform and favourable.
- ISUP 2019 vs GUPS 2019 rules on IDC-P and targeted cores (table below).
- **Biopsy (2005) score = commonest plus highest pattern**; a single pattern is doubled. The lowest score in practice is **6**, which has **no nodal metastatic capacity**, and EPE is uncommon.

Pattern	Morphology
3	Discrete well-formed glands
4	**Cribriform (incl. glomeruloid)**, poorly formed or fused glands
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

GG (International Society of Urological Pathology (ISUP); World Health Organization 2016)	Gleason	Nodes at RP	5-yr BCR-free
1	3+3	0%	**96%**
2	3+4	0.6%	**88%**
3	4+3	4.3%	**63%**
4	4+4, 3+5, 5+3	—	**48%**
5	9–10	—	**26%**

KEY POINT

The steep drop is between GG2 and GG3: never lump them as "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- Report **% pattern 4** for GG2–3† (ISUP 2014). GG2 with limited pattern 4 may suit AS. GG3 with 90% pattern 4 needs neoadjuvant hormones with radiotherapy (RT); at 60% possibly not.
- **Tertiary pattern 5** (<5% at RP) raises BCR risk but **stays in its GG**, because nodal risk is not raised.

Issue	ISUP 2019	Genitourinary Pathology Society (GUPS) 2019
% pattern 4 in GG2–3; cribriform	†Report	†Report
IDC-P with invasive cancer	†**Incorporated into grade**	†**Not graded**; comment
Targeted cores of one lesion	†**Aggregate grade** (plus per-core)	†Each core; aggregate optional

- †On biopsy any higher pattern counts as secondary, however small; at RP a higher pattern <5% is tertiary, GG unchanged. Cribriform is pattern 4 whatever its extent.

> **KEY POINT**
> †Cribriform pattern and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under-grading (one core) and over-grading (highest core).

Upgrading setting	Risk
Biopsy GG1, then RP	~⅓ upgraded
Biopsy GG2, then RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Predictors: greater biopsy extent, higher PSA, smaller gland, fewer cores. Reclassification within 3 yr means missed disease, not progression.

4 · Needle biopsy assessment

Why this matters: core handling decides whether cancer is found and graded right.

- Use **separate jars per site** and ≤2–3 cores per block. Pitfalls: **at the base, SV/central zone mimic HGPIN; at the apex, Cowper gland mimics cancer**.
- Report the **number of positive cores plus one other extent measure**; none is superior.
- Report **perineural invasion** on biopsy (raises EPE risk, adverse on AS, prognostic for external-beam radiotherapy); omit it at RP.
- Basal cell markers, **high-molecular-weight cytokeratin (HMWCK) and p63**, stain benign glands, so their **absence supports cancer** (pitfalls: atrophy, adenosis). **α-methylacyl-CoA racemase (AMACR)** is an adjunct only.
- **Adenosis**, in the TZ, is the chief mimic of low-grade cancer and carries **no increased cancer risk**.
- **Atypical small acinar proliferation (ASAP)**: ~5% of biopsies; cancer follows in ~**40%** (PSA does not stratify), ~80% GG1 and ~5% GG4–5. Get **expert review before re-biopsy**, then re-biopsy the atypical site.

> **KEY POINT**
> Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, and an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

Why this matters: the RP report sets recurrence risk and adjuvant therapy.

- At TUR, 6–8 cassettes detect all T1b cancers. **Grade each major nodule**.
- **Nodes** are uncommon with Gleason <8. Node-positive: 15-yr BCR-free ~7%, cancer-specific survival ~58%. Intraductal or cribriform patterns raise metastasis risk.
- **EPE**: focal vs nonfocal extent predicts progression. **SVI** carries **65%** 5-yr progression.
- **Margins** are positive in ~15% (robotic); only ~50% progress. Distance to ink is irrelevant. Report **extent (focal vs nonfocal, 3 mm) and grade at the margin**.
- An **intraprostatic incision** carries the same risk as focal EPE with a positive margin.
- Independent predictors of BCR are grade, EPE extent, margins, SVI and nodes. **Tumour volume is not independent**.

6 · Treatment effect

Why this matters: grading treated tissue can mislead.

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade looks artefactually higher	**No** where effect is present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no treatment effect
Cryotherapy / high-intensity focused ultrasound (HIFU)	Infarction; residual cancer unaltered	**Yes**

Post-RT biopsy: cancer with treatment effect has a prognosis close to no cancer.

7 · Subtypes and other tumours

Why this matters: some subtypes behave and stage differently.

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at the verumontanum, mimics papillary urothelial cancer; PSA may be normal; behaves like Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in the prostate** (35–45% of cystoprostatectomies): if only intraductal, bladder stage decides; from urethral carcinoma in situ, suburethral invasion is pT1 and stromal pT2. **Transmural spread from the bladder is pT4**.
- Immunohistochemistry: prostate stains PSA, P501S, NKX3.1; urothelial **GATA3**.

- Sarcomas: **rhabdomyosarcoma in children, leiomyosarcoma in adults**. Stromal tumour of uncertain malignant potential (STUMP) may recur: follow up.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

Why this matters: curable cancer is silent.

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced** disease causes voiding obstruction, ureteric obstruction (loin pain, renal failure), haematuria and reduced ejaculate.
- **Metastatic** disease causes bone pain, fracture, leg oedema and cord compression; rarely retroperitoneal fibrosis, paraneoplastic syndromes or disseminated intravascular coagulation (DIC).

2 · Screening in brief

Why this matters: screening is how most cancers are now found (details in Ch 153).

- P2 PSA screening for prostate cancer: evidence, problems and solutions — *ERSPC vs PLCO, NNI 570 / NND 18, USPSTF D to C, AUA, NCCN, overdiagnosis*

3 · DRE and PSA

Why this matters: these two tests decide who goes on to imaging and biopsy.

- **Digital rectal examination (DRE)** has low screening value at PSA <3. But an abnormal DRE raises the risk of Grade Group (GG) 4–5 and multiplies risk at every PSA level. DRE barely changes PSA.
- **Prostate-specific antigen (PSA)** is organ-specific, not cancer-specific. It rises ≈4% per mL of volume, and serum PSA without cancer comes mainly from the transition zone (TZ).
- PSA is lowered by hypogonadism, obesity, statins, 5α-reductase inhibitors (5-ARIs) and benign prostatic hyperplasia (BPH) surgery. On a 5-ARI (finasteride 1 or 5 mg, dutasteride), **double PSA** after ≥12 months, or treat the nadir as the new baseline.
- Risk is **continuous**: the Prostate Cancer Prevention Trial (PCPT) found cancer in 15% of men with PSA ≤4, and ≈15% of these were Gleason ≥7.
- Thresholds are 4 or 2.5–3 (the European Randomized Study of Screening for Prostate Cancer (ERSPC) used 3); 10 has been suggested above 70 y. Repeat a raised PSA before biopsy, and give no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death >25 years later (Malmö).
- Derivatives: PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (at PSA 4–10). *See Ch 154, 155*

4 · Staging: TNM (AJCC 8th, 2017)

Why this matters: stage sets the treatment options and the prognosis.

Pathological stage (radical prostatectomy (RP) specimen and nodes) is more accurate and prognostic than clinical stage (DRE, PSA, biopsy, imaging).

Category	Definition
T1a / T1b	Incidental at transurethral resection (TUR): ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ of one side / >½ of one side / both sides
T3a / T3b	Extraprostatic extension (EPE) / seminal vesicle invasion (SVI)
T4	Fixed, or invading sphincter, rectum, bladder, levators or pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites with or without bone

American Joint Committee on Cancer (AJCC) prognostic groups add PSA and GG. Whitmore–Jewett staging is historical.

> **KEY POINT**
> T1 is defined by how the cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; a biopsy-found impalpable cancer is T1c even if bilateral; clinical T is set by DRE, not imaging.

5 · Predicting extent and risk

Why this matters: risk groups, not single tests, guide treatment.

- **PSA** cannot stage an individual. Organ-confined: ≈80% at PSA <4, 66% at 4–10, <50% above 10. Nodes: ≈20% at PSA >20 and **75% at >50**.
- **Biopsy grade** is the most important biopsy finding. *See Ch 156*
- Tertiary pattern 5 on biopsy is recorded as the secondary pattern, so 3+4 with tertiary 5 becomes 3+5. Tumour in the seminal vesicle or periprostatic fat on biopsy worsens prognosis. Acid phosphatase has been abandoned.

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all criteria)	T1–T2a	≤10	≤6	83%
Intermediate (any one)	T2b	>10–20	7	46%
High (any one)	T2c	>20	8–10	29%

† National Comprehensive Cancer Network (NCCN) group	Criteria
Very low	T1c, GG1, PSA <10, PSA density <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- The Cancer of the Prostate Risk Assessment (**CAPRA**) score and **nomograms** predict better than groups for individuals (†Partin tables, MSKCC). Nodal nomograms decide pelvic lymph node dissection (PLND).
- **Tissue genomic tests** (Prolaris, Oncotype DX 17 genes, Decipher 22 RNA) refine risk, but prospective validation is awaited.

6 · Staging imaging and nodes

Why this matters: the risk group chooses the scan.

Risk	Imaging
Low / intermediate, asymptomatic	No routine computed tomography (CT) or bone scan
High	Bone scan with CT or mpMRI; consider PSMA PET

- **Multiparametric magnetic resonance imaging (mpMRI)** is used for local staging (EPE, SVI), treatment planning and active surveillance selection.
- **Prostate-specific membrane antigen positron emission tomography (PSMA PET)**: ^{68}Ga-PSMA-11 and piflufolastat F-18 are Food and Drug Administration (FDA)-approved for high-risk staging. It is more sensitive and specific for nodes and metastases; benefit when conventional imaging is negative is unproven.
- **PLND** has no consistent oncological benefit; its value is **staging**. Do it when the nomogram nodal risk exceeds the guideline threshold.

KEY POINT

Imaging yield follows pre-test probability. In low-risk men most positive scans are false positives, which lead to more tests, anxiety and wrongly denied cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening randomised trial, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at the cost of overdiagnosis
PLCO	US screening randomised trial, from 1993	No mortality difference	Contaminated control arm; ≈3× fewer events
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
† proPSMA (2020)	PSMA PET-CT vs CT and bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET is the better first-line staging test

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

Why this matters: most localised cancers will not kill the man, so the first question is whether and when to treat.

- In 15-year randomised data on favourable risk, surveillance gave **more metastasis but no difference in cancer mortality**, partly because many crossed over to treatment.
- **Spectrum**: a man likely to die of competing risks gets **watchful waiting (WW)**. Good longevity with unclear benefit means **active surveillance (AS)**.
- A man who needs therapy gets radical prostatectomy (RP), external-beam radiation therapy (EBRT), brachytherapy, or whole/partial-gland ablation.

KEY POINT
Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act *before* the cancer does.

2 · Host evaluation

Why this matters: life expectancy decides whether treatment can help; baseline function decides which treatment hurts least.

- The National Comprehensive Cancer Network (NCCN) asks for life expectancy from Social Security Administration (SSA) or World Health Organization (WHO) tables.
- In the Prostate Cancer Outcomes Study (PCOS), 10-year prostate cancer mortality was **3–18%** by risk, but other-cause mortality was **≥33%** with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men, so individualise. **Age bias**: clinicians judge by age, not comorbidity.
- The **Charlson index (CCI)** was designed for 1-year inpatient mortality; **PCCI** reweights it for 10-year mortality.
- The automated **Care Assessment Needs (CAN) score** (Veterans Health Administration) best predicts 5-year survival MCQ.
- The number needed to treat (NNT) rises steeply with Charlson score.

Baseline function (International Prostate Symptom Score [IPSS], UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive lower urinary tract symptoms (LUTS), large gland	Surgery	RP relieves obstruction; brachytherapy increases retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm^3	Surgery (or neoadjuvant androgen-deprivation therapy, ADT)	Pubic arch interference; if >80 g, ADT (shrinks 25–40%) before radiotherapy (RT)
Chronic retention / neurogenic bladder	Surgery	RT causes bladder fibrosis; if RT is chosen, switch to intermittent self-catheterisation

Baseline state	Favours	Reason
Prior transurethral resection (TUR) defect	Surgery	Brachytherapy dosimetry is hard; incontinence increases. Do TURP *before* RT and delay EBRT **6–8 weeks**
Inflammatory bowel disease (IBD), bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

- **Sexual** recovery depends on age, baseline erection, PDE5 inhibitor use and vascular risk. Poor baseline function means non-nerve-sparing surgery is used more.

Modality	Absolute contraindication	Relative
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

Why this matters: the risk group sets the default advice and whether staging scans are needed.

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; grade group (GG) 1; prostate-specific antigen (PSA) <10; <3 positive cores, ≤50% each; PSA density < **0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	
Favourable intermediate	Intermediate-risk factors (IRFs): T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan and CT/MRI, plus chest imaging; prostate-specific membrane antigen (PSMA) PET may replace these (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

KEY POINT

Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% of any core.
- Nomograms: **Partin** predicts pathological stage (accurate 72.4%); **Kattan** predicts relapse after RP; with **UCSF-CAPRA**, risk jumps at score ≥6.
- **proPSMA** (302 high-risk men): PSMA PET had accuracy **92% vs 65%** and sensitivity 85% vs 38% over conventional imaging. A negative scan cannot exclude small nodes.
- Genomic tests are not routine (AUA 2022). Use them only where they change an AS decision (high-volume 3+3, or 3+4 considering AS).

4 · Observation vs active surveillance

Why this matters: both mean "no treatment now", but AS aims at later cure and WW only at comfort.

- P2 Expectant treatment of CA prostate — *AS (curative, deferred) vs WW (palliative): candidate, monitoring, triggers*

Not in the paper answer:

- Systematic biopsy misses high grade in 25–45%; most upgrading is to GG2, which has low metastatic potential in the first decade.
- PSA density >0.15 and core cancer density predict reclassification.
- SPCG-4 relative risk 0.56, NNT 8; PIVOT benefit only in intermediate risk; ProtecT treated 60% of monitored men by 15 years.
- AS is **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 is acceptable.
- Limitations are **25–45%** occult high grade on systematic biopsy, declining adherence and no agreed triggers. *See Ch 159*

5 · Comparative oncological evidence

Why this matters: observational data seem to favour surgery, but the gap is mostly bias.

- Biochemical-free survival is a flawed RP-vs-RT end point, because RT leaves PSA-producing tissue.
- Population studies favour RP (Swedish subdistribution HR ≈1.8; PCOS disease-specific HR 0.35, curves apart by 3 years). This is largely **selection bias**.

6 · Landmark trials

Why this matters: these three trials underpin every treat-or-observe discussion.

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA; ~90% palpable	23.2 yr: cancer deaths 63 vs 99, relative risk (RR) **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger men with intermediate risk
PIVOT	RP vs observation; 731 (2000 planned); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause absolute risk reduction (ARR) 5.5% (not significant), cancer 4.0% (not significant); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT with 3–6 mo ADT; 77% low risk	15-yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP equals RT

KEY POINT

Trials show little benefit because competing mortality and low-risk enrolment dilute the effect; benefit appears only in younger intermediate-risk men with long follow-up. ProtecT expected 15% mortality but saw ~1% at 10 years.

7 · Functional outcomes compared

Why this matters: cancer control is similar, so side effects usually decide.

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**; persists at 24 mo	No change

- **PCOS** found differences at 2 and 5 years but **none at 15 years**.
- **ProtecT** found rising bloody stools after RT, and no difference in anxiety, depression or quality of life.
- In **CEASAR**, the erectile disadvantage of RP was confined to the top quartile of baseline function.
- One trial of open vs robotic RP found similar function at 12 weeks; the surgeon matters, not the approach. *See Ch 160/161*

KEY POINT

Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years, so the choice is *when and where* the patient accepts harm.

8 · Shared decision-making

Why this matters: with no clear winner, counselling shapes the choice.

- No treatment is clearly superior, so **patient preference carries unusual weight** (AUA, ASTRO, SUO).
- The **urologist's recommendation is the most influential factor**, so beware your own bias.
- **Longevity assessment may be the most important part of counselling**.

9 · Stepwise decision algorithm

Why this matters: four questions in order keep the consultation clear.

ALGORITHM

1. **Treat or not?** Weigh disease (grade, stage, genomics, family history) and host (age, comorbidity, longevity, preference). If longevity is short, then WW; if very low/low or selected favourable intermediate, then AS.
2. **Partial or whole gland?** Weigh focality, size, grade, potency, LUTS. If limited-focality intermediate risk, then focal ablation, ideally in a trial. *See Ch 163*
3. **Surgery, radiation or whole-gland ablation?** Compare outcomes, toxicities, baseline function and risk of secondary treatment.
4. **Delivery:** open vs robotic RP, node dissection, surgeon; or EBRT vs brachytherapy, fractionation, ADT. *See Ch 162*

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies are options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP, or RT with ADT; AS strongly discouraged
High / very high	RP, or RT with ADT; consider even with competing risks if the cancer threatens within 10 yr; no focal therapy

- **Surgery:** early incontinence and erectile dysfunction (ED) are the norm, so set a realistic timeline and quote your own outcomes. LUTS may improve. Men <65 had lower cancer mortality (ProtecT).
- **Radiation:** it spares short-term function but causes yearly decline. Secondary rectal/bladder cancer may follow at 15–20 y (controversial). ADT for intermediate/high risk drives early toxicity.
- **Whole-gland ablation:** long-term data are scant, and fibrosis hampers salvage surgery. Cryoablation almost always causes ED. HIFU (high-intensity focused ultrasound) preserves function better, but retreatment is **29–38%**.
- EBRT, brachytherapy, SBRT and protons give comparable 10-year cancer control.

10 · MCQ one-liners

MCQ ONE-LINERS

1. Early divergence of RP vs RT survival curves in observational studies indicates selection bias.
2. Brachytherapy, not EBRT, causes the worst acute obstructive/irritative symptoms.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

Why this matters: most screen-detected cancers would never harm the man.

- **Latent reservoir**: autopsy cancer prevalence roughly equals age as a percentage. Prostate-specific antigen (PSA) testing and biopsy detect cancer in 15–25%, yet lifetime risk of cancer death is only **3%**.
- Harbouring : death is ≈20:1 and diagnosis : death ≈6:1, so overdiagnosis is 87–94%.
- Active surveillance (AS) rests on **two premises**: progression takes a long time, and cure is still possible at the first sign of higher risk.

2 · AS versus watchful waiting

Why this matters: one aims at cure, the other does not.

- P2 Expectant treatment of CA prostate — *AS curative/deferred vs WW palliative; candidates, monitoring, switch at life expectancy <5–10 yr*

Not in the paper answer:

- The switch to watchful waiting (WW) is typically near age 80 (age–comorbidity nomograms).
- Older observation trials and registries reflect WW, not modern AS.
- The US Preventive Services Task Force moved PSA screening from Grade D (2012) to C (2018) because AS cut overtreatment.
- **Pitfall — neuroendocrine dedifferentiation**: elderly men with unrecognised hypogonadism may progress without a PSA rise. So measure testosterone and get interval imaging.

> **KEY POINT**
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

Why this matters: AS is safe only because grade group (GG) 1 cancer almost never spreads.

- **Pattern 3 is molecularly near-normal.** Patterns 4–5 show EGFR overexpression, HER2 amplification, BCL2, angiogenesis and **PTEN deletion in 70–90%**; cyclin D2 is lost in pattern 4.
- **MYC amplification with PTEN deletion** drives metastasis.
- **Metastatic potential of pure GG1 is effectively zero**, though occasional extraprostatic extension (EPE) occurs.
- True grade progression is ~**1–2% per year**, mostly in high-volume Gleason 6, so follow-up must be long term.
- Deaths in low-risk series come from **occult higher grade** missed at biopsy, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

KEY POINT

The main hazard of AS is **sampling error at entry**, not biology over time ("wolf in sheep's clothing"). First job: get the entry grade right.

4 · Patient selection

Why this matters: who enters AS sets the cancer death rate.

Most GG1 should be on AS (AUA, EAU, ASCO, NICE and Cancer Care Ontario agree).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher
Metastasis	Exceptionally low	Exceptionally low

The split reflects the chance of **occult higher grade**. **High-volume GG1** is still a candidate, but check it with multiparametric MRI (mpMRI) and targeted biopsy, re-biopsy or a genomic assay.

- **Toronto** is inclusive (all Gleason 6 and selected intermediate; ~50% eligible). Its 15-yr cancer mortality is **5%** actuarial (1.5% actual).
- **Johns Hopkins** is restrictive (Epstein very low; ~20% eligible), with 15-yr cancer mortality of **0.5%**.
- So **inclusion criteria set mortality, while intervention stays constant (~25% at 5 yr)**.
- In Toronto, baseline Gleason 7 had ≥20% metastasis at 15 yr (hazard ratio [HR] 3.75); PSA >10 barely predicted it.
- In PRIAS (n=5302), cancer mortality was <1% at 10 yr. Other-cause mortality far exceeds it; cardiovascular disease is the commonest cause of death.
- **Favourable intermediate**: offer AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk.
- Exclude GG ≥3; intraductal or cribriform histology raises concern.
- **Black men** are AS candidates despite inconsistent data, perhaps with closer surveillance.
- **Family history** rarely justifies treatment ("family history inflation"). The exception is metastatic disease <60 yr in a first-degree relative, which calls for germline testing.
- **BRCA2** carries the highest upgrading risk (to GG ≥3 within 10 yr); BRCA1 and ATM less. Consider definitive therapy even for low grade, or AS only with very close follow-up.
- Risk factors for higher grade are **PSA density >0.15**, Black race and biopsy volume. Restricted diffusion on MRI is a powerful predictor.

5 · Monitoring protocol

Why this matters: AS is safe only with scheduled follow-up.

ALGORITHM

1. At diagnosis, obtain mpMRI.
2. Do a **confirmation biopsy** (systematic + targeted) within 1–3 yr.
3. Check PSA 6-monthly, digital rectal examination (DRE) every 1–2 yr and MRI ≤ every 18–24 months.
4. Repeat biopsy every 3–5 yr while stable.
5. If a trigger appears, then give definitive therapy; if life expectancy falls, then switch to WW.

- Confirmation biopsy may be omitted if the first biopsy was MRI-targeted; delay to year 3 is harmless (modelling).
- **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly in year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume: get MRI or a biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2: may continue AS
	PSA rise alone: repeat MRI or biopsy

KEY POINT

Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

Why this matters: MRI is strong at entry but weak as the only follow-up test.

- Diffusion-weighted imaging (DWI) signal is inversely related to Gleason, so MRI misses low grade but locates higher grade.
- **At entry the evidence is strong**: mpMRI before confirmation biopsy, with **systematic and targeted cores**, upgrades about ⅓ (found by both routes). NICE recommends it; NCCN lists it as an option.
- **Serial MRI alone is weak** (negative predictive value [NPV] ~0.8–0.9 for GG ≥2), so it must never be the only way to exclude progression.
- MRI-invisible cancers behave better, and a visible lesion predicts treatment. Higher-risk men still need systematic biopsy.

7 · Biomarkers and genomics

Why this matters: in the wrong men these tests mislead.

- Decipher (22-marker RNA) predicts upgrading in GG1, not GG2. Oncotype DX is mixed (none in Canary PASS). Prolaris has no AS-specific evidence. None was built for AS.
- **Bayesian problem**: with a 3% prior metastasis risk, a 90%-specific test labels 10% "significant". False positives are about 3× true risk, so test only higher-risk candidates (high volume, high PSA density, GG2).
- Use the tests to set surveillance intensity, **never as the sole basis for AS**. A favourable score does not override adverse clinical or pathological features.

8 · Lifestyle, drugs and psychology

Why this matters: some popular supplements are harmful.

- **Stop smoking**, exercise vigorously (ERASE: lowered PSA), keep a healthy weight and eat a vegetable-based diet.
- Only vitamin D 1000–2000 IU/day is reasonable. **Selenium and vitamin E are contraindicated** (SELECT).
- 5α-reductase inhibitors (5ARIs) are not used to prevent progression (REDEEM). Statins are reasonable for cardiovascular disease; metformin, apalutamide and enzalutamide are investigational.

9 · Landmark trials

Why this matters: observation is mortality-safe but not metastasis-free.

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; radical prostatectomy (RP) vs WW; 23 yr	Cancer death 19.6% vs 31.3% (relative risk 0.55)	Benefit at age <65 and GG ≥3; none in GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (not significant); low-risk cancer death 2.7%	Modest benefit in intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs radiotherapy; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring is not AS (no scheduled biopsy, >⅓ intermediate/high)
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (not significant); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

Why this matters: open surgery set the rules robots follow.

- Radical prostatectomy (RP) alone has been shown in a randomised controlled trial (RCT) to reduce metastasis and cancer death (**SPCG-4**).
- Most RPs are now robotic, without proven benefit. *See Ch 161*
- **Goals, in order**: cancer control, then avoiding technical complications, then continence, then potency.

> **KEY POINT**
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

Why this matters: bleeding, margins, continence and potency all depend on anatomy.

- P1 P4 Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — *fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes*

Not in the paper answer:

- Ligating the "lateral pedicle" mid-way cuts both vessels and nerves.
- The growing prostate thins the sphincter over a broad apical attachment.
- An apical vessel tethers the NVB, so divide lateral sphincter edges midway to the pelvic floor.
- Divide the posterior sphincter left, right, then centre; never all from one side.
- The dorsal venous complex (DVC) **common trunk lies directly over the urethra**, where it is transected.
- Inferior vesical urethral branches enter at the bladder neck (**5 and 7 o'clock** pedicles); capsular branches mark the cavernous nerves.
- The pelvic plexus combines parasympathetic **S2–S4** and hypogastric sympathetic fibres. It lies beside the rectum, **5–11 cm** from the anal verge, centred on the seminal vesicle tip.
- At the apex, neurovascular bundle (NVB) branches spray anteriorly and posteriorly, the basis of **high anterior release**.
- The **striated sphincter** is a vertical tubular sheath (Oelrich), not a flat diaphragm, so apical dissection decides continence.
- Sphincter (slow-twitch) gives passive, levator ani active continence. Both are mainly **pudendal**, so pelvic floor exercises train both.

> **KEY POINT**
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

Why this matters: timing makes the planes safer.

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, transurethral resection of the prostate (TURP), hernia mesh, large gland.

- **Defer surgery 6 weeks after biopsy and 12 weeks after TURP**, so haematoma settles.
- Inguinal hernia affects ~15% before and up to 20% after RP, so examine standing and repair it preperitoneally at RP.
- No autologous donation; cephalosporin prophylaxis. The **fibreoptic headlight** is the most useful instrument.

4 · Retropubic RP — operative steps

Why this matters: each step controls bleeding, secures the margin or protects nerves.

ALGORITHM

1. Lower midline extraperitoneal incision; pelvic lymph node dissection (PLND).
2. Open endopelvic fascia; divide puboprostatic ligaments.
3. Control and divide the DVC, then the urethra and posterior sphincter.
4. Release the NVBs; divide pedicles, bladder neck and seminal vesicles.
5. Reconstruct the bladder neck and make the anastomosis.

- **PLND** is for intermediate/high risk, omitted for Grade Group 1–2 low-volume disease. *See Ch 161*
- **Endopelvic fascia**: open at the sidewall reflection; too medial causes Santorini bleeding.
- **Puboprostatic ligaments**: divide, but spare the **pubourethral** part (anchors the sphincter).

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if the catheter moves freely	3-0 figure-of-eight to pubic perichondrium, plus a back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- The **anterior apex is the commonest positive margin site**. Get absolute venous control before touching it: oversew the distal DVC as a "hood".
- **Urethra**: transect close to the apex; six 3-0 sutures, **mucosa and submucosa only** (muscle bites delay continence).

Nerve sparing

- The **interfascial** plane (levator vs prostatic fascia) is correct. Walsh never uses intrafascial (margin risk). Wide (extrafascial) excision is for induration, a fixed bundle or doubtful margin.
- Both NVBs are usually kept: extracapsular extension rarely exceeds 1–2 mm, and margins are equal with or without sparing.
- Decide in theatre: **if induration or a fixed bundle, excise that side** (touch is absent robotically).
- **Technique**: release the prostate from the bundle, not the bundle from the prostate. Start **at the apex**, work to the base, ligate the posterior branch at the seminal vesicle. Small clips; **never thermal energy**.
- **High anterior release** (before urethral division) means less traction and faster recovery. Unilateral equals bilateral, so do the favourable side only.
- Avoid it if ≥2 **Tsuzuki criteria** are present: prostate-specific antigen (PSA) >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal digital rectal

examination (DRE).

- **Wide excision**: clamp **medial to lateral** on the rectum (reverse risks rectal injury).

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; the pelvic plexus lies on their lateral surface. If a posterolateral margin looks doubtful, excise that NVB.
- **Tennis-racquet** closure with mucosal eversion; **intussusception** sutures give the "**turtle sign**".
- Make a tension-free anastomosis; a peritoneal window lets haematoma decompress.

> **KEY POINT**
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

Why this matters: a less-used route that bleeds less.

- Perineal RP reflects the fasciae off the prostate, sparing the DVC, so there is **less bleeding**. Retropubic RP works outside the fasciae and must ligate the DVC.
- † PLND needs a separate approach.

6 · Intraoperative complications

Why this matters: anticipate bleeding and rectal injury.

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose with fine non-absorbable suture
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair and omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss is 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish the prostatectomy, dilate the anus, close in 2 layers and place omentum between rectum and anastomosis to prevent a rectourethral fistula.

7 · Postoperative care

Why this matters: early catheter removal causes retention.

- Discharge day 1–2. Ketorolac if creatinine <1.2 mg/dL and no haematuria.
- Keep the **catheter ~1 week**; day-4 removal causes frequent retention. Never remove it with significant extravasation.
- No imaging if haematocrit falls <6%. Oxybutynin 5 mg for bladder spasm.

8 · Postoperative complications

Why this matters: most are preventable or treatable early.

- Mortality is **0.2%**. **Delayed haemorrhage** needs **early exploration and evacuation**, which shortens stay.
- **Thromboembolism** is the major cause of death. It is highest with PLND and peaks at **14–28 days**. Use compression and early ambulation; treat with low-molecular-weight heparin (LMWH).
- **Bladder neck contracture** appears at ~3 months. Steps: dilatation; cold-knife incision at 3, 6 and 9 o'clock with self-catheterisation; triamcinolone; rarely reconstruction with an artificial sphincter.
- **Lymphocele**: *see Ch 161*.
- P2 Post-prostatectomy incontinence — *rhabdosphincter dependence, evaluation, PFMT, sling vs AUS*

Not in the paper answer:

- Other causes: deep sphincter bites, NVB injury, poor bladder neck. Prior outlet obstruction slows recovery.
- Imipramine or adrenergic agonists may help, by cause.
- **Incontinence** is intrinsic sphincter deficiency from injury at DVC division. Prevent it with a preserved apical sphincter, a precise tension-free mucosal anastomosis and a small supple neck.
- Treat with less fluid, no caffeine, no α-blockers and pelvic floor exercises. Continence means **0–1 pad/24 h**.
- **Erectile dysfunction (ED)**: recovery is predicted by **age <65, preoperative potency, no diabetes and bilateral nerve sparing**. It continues up to 4 years.
- PDE5 inhibitor rehabilitation helps; nightly is not better than on-demand.

> **KEY POINT**
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Why this matters: numbers to quote, with caution.

Time	Continent	Potent (Walsh)
3 months	80%	38%
12 months	93%	73%
18–24 months	97%	86%

- Single-expert figures; elsewhere only a third regained baseline potency by 2 years.
- RP improves lower urinary tract symptoms for up to 15 years.

10 · Landmark trials

Why this matters: they support RP and penile rehabilitation.

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works

Trial	Design	Result	Take-home
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

Why this matters: robotic radical prostatectomy (RP) is now the usual route.

- P1 Open vs laparoscopic vs robotic radical prostatectomy; Robotic radical prostatectomy — *history, comparison table, contraindications, ports, steps, complication rates*

Not in the paper answer:

- Avoid relative contraindications (including neoadjuvant hormones) early in the learning curve.
- Also below: MRI accuracy, no benefit of early "rehabilitative" PDE5 inhibitors, rectal repair without colostomy, salvage outcomes.

> **KEY POINT**
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery. *See Ch 160*

2 · Patient selection

Why this matters: few men are excluded, but some are harder.

- The only absolute contraindications are uncorrectable bleeding diathesis and unfitness for general anaesthesia.
- Use prostate-specific membrane antigen (PSMA) PET-CT if high risk with negative conventional imaging.
- After extraperitoneal mesh repair, use the **transperitoneal** route (Retzius obliterated).
- **Obesity**: place longer trocars more cephalad and lower pneumoperitoneum to 10 mm Hg.
- **Large gland/median lobe**: longer, more blood loss, but comparable continence. **Prior transurethral resection (TURP)** brings the ureteric orifices close to the bladder neck.
- Multiparametric MRI (mpMRI) for extracapsular extension (ECE) has specificity **0.88** and sensitivity **0.61**; it guides nerve sparing.

3 · Preparation and anaesthesia

Why this matters: head-down tilt and CO_2 bring new risks.

- Consent for **conversion to open**; **general anaesthesia is mandatory**.
- **Positioning**: tuck and pad arms; **no fixed shoulder rests** (brachial plexus injury) and avoid exaggerated hip flexion (femoral neuropraxia).
- CO_2 pneumoperitoneum causes vagal response, oliguria and hypercarbia, so raise ventilation and lower pressure.
- Restrict fluids to 1.5–2 L, mostly after the anastomosis.
- **Steep Trendelenburg** causes facial, laryngeal and corneal oedema and raises intraocular pressure. Protect the eyes; consider delayed extubation if the face is very swollen.

4 · Access

Why this matters: the route is mostly preference.

- **Extraperitoneal**: smaller space, bowel protected, less tilt (suits obesity and lung disease), but more CO_2 absorption.
- **Transperitoneal**: larger space; preferred after extraperitoneal mesh. Neither has a consistent advantage.
- Transperitoneal **anterior** (commonest; Retzius first) differs from **retrovesical** (seminal vesicles first, useful with a median lobe).
- Place the camera port **15–17 cm** from the symphysis; insufflate to 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

Why this matters: without touch, visual cues guide each step.

ALGORITHM

1. Drop the bladder (Retzius); open the endopelvic fascia; ligate the DVC.
2. Divide the bladder neck; free the seminal vesicles and vasa.
3. Develop the posterior (Denonvilliers) plane; control pedicles and release the NVBs.
4. At the apex, divide the DVC and urethra; inspect the specimen.
5. Do pelvic lymph node dissection (PLND); entrap the specimen.
6. Posterior reconstruction, anastomosis, extraction.

- Open the endopelvic fascia sharply with **no cautery**. Preserve accessory pudendal arteries.
- **DVC**: pneumoperitoneum tamponades venous bleeding. Suture close to the pubis, staple, or divide late and oversew.

Bladder neck — no tactile feedback

- Cues are the prevesical fat line, balloon traction and the instrument "pinch".
- Divide anteriorly in the midline, lift the catheter, inspect for median lobe and orifices, then divide the posterior neck angled **45°** downward.
- After TURP the balloon falls into the defect, so division drifts too distal. A balloon off the midline suggests a median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

- P1 P4 Anatomy of prostate and NVB; NVB preservation in RP — *nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin*

Not in the paper answer:

- Early NVB release, pedicle clip migration and the apical urethral rim (below).
- Incise Denonvilliers ~0.5 cm below the vesicle bases. Bleeding means too close to the prostate. Mobilise the rectum widely early.
- The **intrafascial** plane is accepted here in selected men (Ch 160 never uses it).
- **Athermal (veil) release**: incise levator fascia anteromedially and sweep the NVB posterolaterally off the white prostatic fascia. Then Hem-o-lok the pedicle and divide sharply.
- Release the NVB **before** mobilising the specimen. NVB bleeding needs fine sutures, not cautery.
- Pedicle clips near the anastomosis can migrate into the bladder, causing haematuria, infection and stones.

Apex and reconstruction

- Preserve urethral length but watch for an apical lip. A small urethral rim on the apex lowers margins without harming continence.
- Reconstruct the bladder neck (tennis racquet) only for a large gland, median lobe or prior TURP. Protect the orifices (methylene blue, stent).
- The **Rocco stitch** joins the Denonvilliers remnant and posterior bladder to the posterior rhabdosphincter. Its continence benefit is inconsistent; its agreed benefit is a **tension-free anastomosis**.
- **Van Velthoven running anastomosis**: two arms run anteriorly from 6 o'clock (now barbed); no eversion sutures are needed.

6 · Continence-preserving variants

Why this matters: earlier continence, at some cost in margins.

- P4 Newer surgical concepts in CA prostate — *Retzius-sparing: earlier continence, more margins; Hood: early data*
- Retzius-sparing leaves the detrusor apron, puboprostatic ligaments and DVC untouched; the anastomosis is done anterior first ("upside-down"). The idea is that preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

Why this matters: PLND is the best nodal staging, with its own morbidity.

- P4 P1 PLND templates; Lymphocele — *templates (extended ~63%), staging not survival; lymphocele drain, then fenestration*

Not in the paper answer:

- AUA 2018 and AUA/ASTRO 2022 indications; nomograms predate PSMA PET; complications ~1 in 7; node count is a poor quality surrogate.
- Super-extended PLND adds presacral nodes. The standard template misses most drainage, so **if PLND is done, use the extended template**; morbidity rises with extent.
- **Indications**: may omit in low risk (nodes <5%). American Urological Association (AUA) 2018: unfavourable intermediate or high risk.
- European Association of Urology (EAU) 2021: nomogram risk **>5%**. AUA/American Society for Radiation Oncology (ASTRO) 2022: risk-stratified by nomogram.
- **Briganti** and MSKCC nomograms predate mpMRI and PSMA PET.
- **Extended boundaries**: proximal 2 cm above common iliac bifurcation (ureter crossing); distal node of Cloquet; lateral **genitofemoral nerve**; medial bladder wall. Take all obturator and internal iliac tissue.
- **Clipping lymphatics** means fewer lymphoceles. Open and laparoscopic PLND are equivalent; node count is a poor quality surrogate.

PLND complications (~1 in 7)

- Obturator nerve: remove a clip; repair a transection primarily. The ureter is at risk proximally, so minimise heat there.
- **Lymphocele is commonest** (**2–5%** significant). **Transperitoneal access is not protective**, because the bladder walls off the fossa. A **peritoneal interposition flap** reduces it. It classically presents with ipsilateral deep vein thrombosis (DVT).

8 · Postoperative care

Why this matters: catheter timing drives retention.

- Enhanced Recovery After Surgery (ERAS); same-day discharge for selected men. A drain is not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal at day 3–4 causes more retention. Extravasation delays but does not reduce long-term continence.
- Pelvic floor training starts preoperatively and ≥3 days after catheter removal.
- **Penile rehabilitation**: phosphodiesterase-5 inhibitors (PDE5i) help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Use injections for non-responders; a vacuum device preserves length.

9 · MIS-specific complications

Why this matters: minimally invasive surgery (MIS) has fewer, different complications.

Complication	Rate	Management
Transfusion	1–2%	Low, because pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation or incision; rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors
Robotic failure	<0.5%	Cancel or convert
Recognised rectal injury	Rare; higher in salvage	Multilayer repair and flap, no colostomy
Unrecognised rectal injury	Rare	Presents as rectovesical fistula; colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Small bowel injury** usually happens off-screen; delayed peritonitis needs urgent repair.
- **Ureteric injury** sites: behind the vas (posterior approach); posterior bladder neck after TURP or median lobe; proximal PLND.
- DVT: image the pelvis for lymphocele or haematoma. Significant lymphocele: duplex for occult DVT.

> **KEY POINT**
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

Why this matters: figures for counselling.

- **Oncologic**: positive surgical margin (PSM) rates are equal open and robotic (~15%; ~9% organ-confined). PSM depends on tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume, **not** on obesity, gland size or prior surgery.
- Biochemical recurrence (BCR) means PSA >0.2 ng/mL.
- **Continence**: no-pad rates are in the mid-90s% at 1 year. Persistent mild leakage needs a sling or ProACT; severe (<2%) an artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases (PSMA PET); full bowel prep; air-bubble test for rectal injury; catheter 2–3 weeks. Continence is only 33–80%; most are impotent.

11 · Landmark trials

Why this matters: they frame the open vs robotic debate.

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

- P2 P4 Radiotherapy and brachytherapy for CA prostate; Role of radiotherapy in focal and advanced CA prostate — *α/β, schedules, brachytherapy, ADT by risk, salvage, Phoenix, toxicity*

Not in the paper answer:

- Dose escalation doubled late toxicity without survival gain (§3).
- A single 19 Gy high-dose-rate fraction fails (§5).
- RADAR and RTOG 99-10 on ADT duration (§6).
- Why nodal trials failed (§7).
- Poor-prognosis markers, bounce rules, biopsy timing (§9).
- Salvage prostatectomy incontinence ~63% (§11).

1 · Radiobiology and fractionation

This explains why modern radiotherapy (RT) courses are short.

- Conventional fractionation gives **1.8–2 gray (Gy)** daily.
- In the **linear–quadratic model**, the **α/β ratio** sets a tissue's response to fraction size: low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer has few cycling cells, so its α/β is **≈1.5 Gy** (1.0–1.7 in external beam radiotherapy [EBRT] cohorts), **lower than the rectum (~3 Gy)**. So fractions >2 Gy favour tumour kill over late toxicity: the rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy in 39 fractions)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = stereotactic body RT (SBRT)	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

Aim: more dose to prostate, less to rectum and bladder.

- Multiparametric MRI (mpMRI) is fused with planning CT; MRI alone lacks electron density.
- Prostate-specific membrane antigen PET (PSMA PET) before RT for unfavourable intermediate/high risk changes management in up to half *(see Ch 152)*.
- **Intensity-modulated RT (IMRT) or volumetric modulated arc therapy (VMAT)** uses inverse planning. Versus 3D conformal RT (3D-CRT) it lowers gastrointestinal (GI) but not genitourinary (GU) toxicity, as the urethra lies inside the target. It is standard without a randomised trial.

Volume	Prostate application
Gross tumour volume (GTV)	Dominant intraprostatic lesion (DIL) on MRI
Clinical target volume (CTV)	Prostate, with or without proximal seminal vesicles and pelvic nodes
Planning target volume (PTV)	CTV plus a margin for motion and set-up: **3–7 mm** with image guidance

- **Image-guided RT (IGRT)** with daily gold fiducials or cone-beam CT is the National Comprehensive Cancer Network (NCCN) standard; its smaller margins make large fractions safe.

- A **rectal spacer** (hydrogel or hyaluronic acid) is injected transperineally into Denonvilliers' fascia. Late rectal toxicity: **2% vs 7%** (hydrogel, 79.2 Gy); grade ≥2: 2.9% vs 13.8% (hyaluronic acid, 60 Gy/20).
- The **MR-linac** (MIRAGE trial) allows 2 mm margins, lowering acute GU toxicity.
- **Protons**: the Bragg peak shrinks only the low-dose bath, not the high-dose rectal volume; no randomised benefit over IMRT.

3 · Dose escalation

Higher dose buys better prostate-specific antigen (PSA) control, not survival.

- Six randomised trials (MD Anderson, PROG 95-09, Dutch, MRC RT01, GETUG 06, RTOG 0126) raised **biochemical control in all risk groups**, with **no overall survival (OS) gain**.
- OS did not improve because salvage works and metastasis fell only from 6% to 4%. Still, ≥~78 Gy is standard.
- Cost: grade ≥2 late toxicity doubled at 78 Gy; rectal risk rises when >25% of rectum gets 70 Gy.
- **FLAME**: a focal boost to the MRI DIL improved biochemical progression-free survival (hazard ratio [HR] 0.45) without more toxicity.

4 · Hypofractionation

Shorter courses are now standard; know the proven schedules.

- **Moderate**: CHHiP 57 Gy/19 was not shown noninferior; PROFIT 60/20 was noninferior; RTOG 0415 70 Gy/28 was noninferior but more toxic late; HYPRO 64.6 Gy/19 was **not superior**.
- After prostatectomy, HYPORT 62.5 Gy/25 was noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC (no androgen deprivation therapy [ADT]) gave ~84% 5-yr failure-free survival, with more acute toxicity.
- **Dose ceiling**: no gain above **40 Gy/5**. Give on alternate days with 3–5 mm margins. Rectal risk rises if >35% of the circumference gets >39 Gy; a spacer reduces ulcers.

5 · Brachytherapy (BT)

Implanted sources give a dose EBRT cannot; urinary function decides eligibility.

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV) in a remote afterloader
Technique	Transrectal ultrasound (TRUS)-guided transperineal template; **peripheral loading** spares urethra	Dose shaped by dwell times and positions; no post-implant dosimetry
Uses	Monotherapy for low/intermediate risk; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{103}Pd	17 d	110–125 Gy	90–100 Gy
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- A shorter half-life delivers dose faster, so the prescription is lower.
- **Post-implant dosimetry** (CT with or without MRI, 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. A low D90 predicts biochemical failure. MCQ D90 is a dose, V100 a volume.
- **Relative contraindications**: **International Prostate Symptom Score (IPSS) >20**; **maximum flow (Qmax) <10 mL/s**; large median lobe; large transurethral resection (TURP) defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel or connective tissue disease; **pubic arch interference**; no lithotomy.
- A large gland shrinks ~30% with ADT and ~20% with a 5α-reductase inhibitor (5ARI). Retention risk is ≈29% if IPSS >20 vs ≈2% if <10; start α-blockers before the implant.
- **With EBRT**: 45 Gy plus an implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone equalled EBRT plus BT in intermediate risk with less toxicity, so BT alone is standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU toxicity **18.4% vs 5.2%**.
- UK HDR boost trial: adding HDR 17 Gy/2 to 55 Gy gave a 21% absolute relapse-free gain at 12 yr.

> **KEY POINT**
> Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

ADT duration is chosen by risk group.

- **Synergy**: ADT causes apoptosis, shrinks the target and lowers **Ku70**, impairing non-homologous end-joining repair. RT up-regulates androgen receptor signalling, which ADT blocks *(see Ch 164)*.

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 mo better than 3); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 mo better than 6); PCS IV: 18 mo = 36 mo

- Needed duration falls as RT dose rises: in **TROG 03.04 RADAR**, 18 mo beat 6 mo only without an HDR boost.
- **STAMPEDE**: adding 2 yr abiraterone in high-risk N0/N1 M0 improved metastasis-free survival (MFS) and OS.

7 · Pelvic nodal RT

Older nodal trials failed mainly because the field was too low.

- **RTOG 94-13** (node risk >15%): early progression-free survival gain later lost; **GETUG-01**: none.
- The reason: a field border at L5/S1 misses the **common iliac nodes**, the site of over half of first nodal recurrences. RTOG 09-24 therefore uses **L4/L5**.
- † **POP-RT** (node risk >20%, long ADT): whole-pelvis RT to L4/L5 raised 5-yr biochemical failure-free (~95% vs ~81%) and disease-free survival; no OS difference; more acute GI toxicity.

8 · Post-prostatectomy RT

After radical prostatectomy (RP), most men can wait for a PSA rise.

- RADICALS-RT, GETUG-AFU 17 and RAVES (64–66 Gy), with their meta-analysis, make **early salvage preferred**. Adjuvant RT may still suit very high risk (Gleason 8–10, pT3b) *(see Ch 164)*.
- Dose is ≥64 Gy; RTOG 96-01 grade ≥2 GU toxicity was 14%. PSMA, fluciclovine or choline PET localises recurrence.

9 · Follow-up and biochemical failure

Do not mistake a benign bounce for failure.

- Benign glands persist, so PSA stays detectable. Nadir: ~0.6 ng/mL (70 Gy), ~0.3 (79 Gy), ≤0.1 (EBRT plus BT); reached at ~1.5 yr after EBRT, ~2.5 yr after BT.

	ASTRO 1996	Phoenix 2005 (current)
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated to midway between nadir and first rise	When reached (or when salvage started)
Notes	Delays diagnosis ≥18 mo; invalid with ADT or BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis: failure <18 mo; nadir <12 mo; nadir >2 ng/mL at 1 yr; **PSA doubling time (PSADT) <3 mo** predicts cancer death (if <12 mo, consider early intervention). Local failure rises slowly (PSADT ≥1 yr); distant failure has PSADT 3–6 mo.
- A **bounce** (rise >0.2 ng/mL, then durable fall) occurs in ~1/3 after LDR (2/3 in young men), lasts up to 12–18 mo, mostly in the first 3 yr, and predicts a *better* outcome.
- After BT, watch a rise within 30 mo; biopsy if it persists >30 mo; stage systemically as PSA nears 10 ng/mL.
- **Biopsy** is best at **30–36 mo**, since irradiated cells die slowly. Do not grade marked radiation effect; it behaves as negative. High-molecular-weight cytokeratin separates radiation atypia from cancer. Recurrence arises at the original dominant nodule.

10 · Toxicity

Counsel on bladder, bowel and erections.

Toxicity	Features	Management / prevention
Acute GU	Peaks ~2 mo, settles by 12 mo; after BT the IPSS curve follows the isotope half-life	α-blockers; catheter for retention

Toxicity	Features	Management / prevention
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU toxicity; peripheral loading
Late GI (proctitis)	Bleeding at 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid or sucralfate enemas, then **argon plasma coagulation** or hyperbaric oxygen, then colostomy
Erectile dysfunction (ED)	Vascular, over 24 mo; mostly due to ADT; penile bulb dose; best with BT alone (~76% potent at 5 yr)	Phosphodiesterase-5 inhibitors (PDE5i) work in ~¾
Second cancers	† Small excess of bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

Local salvage helps only truly confined disease.

- Salvage needs **biopsy-proven, local-only** recurrence. Options: salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy and high-intensity focused ultrasound (HIFU).

12 · Landmark trials

These trials underpin the choices above.

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% vs 80%; OS same	Dose escalation improves control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (plus 12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT with no, 3 or 6 mo ADT	6 mo improved OS; 3 mo did not	≥6 mo short-course
RTOG 94-08	66.6 Gy with or without 4 mo ADT, T1b–T2b	10-yr OS 57% vs 62%; intermediate risk only	ADT for intermediate, not low
EORTC 22863	RT with or without 3 yr goserelin, T3–T4	10-yr OS 39.8% vs 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	RT lowered cancer death and raised OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate-only RT	No durable benefit	Field height matters

Trial	Design	Result	Take-home
SPPORT	Salvage bed RT, with or without ADT and nodes	Pelvic RT plus ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

Focal therapy treats only the dangerous tumour, trading some cancer risk for function.

- **Focal therapy (FT)** is guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**. Sparing the neurovascular bundles (NVBs), sphincter and rectum preserves sexual and urinary function.
- The model is **breast lumpectomy**. FT rests on multiparametric MRI (mpMRI), targeted biopsy and the fact that **Gleason pattern 4/5** is the lethal subgroup.

Index lesion hypothesis

- Cancer is multifocal in **50–91%** of radical prostatectomy (RP) specimens. The **index lesion (IL)**, the largest or highest-grade focus, is held to drive natural history and metastasis.
- **For**: the IL holds **80–90%** of tumour volume, and satellites are mostly **<0.5 mL** (Stamey "insignificant"). So 52–93% of glands are functionally unifocal.
- **Against**: a satellite is the aggressive focus in 12–38%; IL Gleason matches overall grade in only ~⅔; a lethal clone was traced to a small pattern 3 focus.

> **KEY POINT**
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT and active surveillance (AS)** form one strategy: ablate grade 4 disease, watch the low-grade remainder, and retreat new foci.

2 · Patient selection

Selection is the core of FT: find, grade and map the lesion.

- The patient must be **compliant with close surveillance**, as recurrence is likelier than after radical therapy.
- **Transrectal ultrasound (TRUS) biopsy** misses almost half the disease relevant to hemiablation compared with **transperineal mapping biopsy (TMB, 5-mm grid)**. Over half of "TRUS-unilateral" men are disqualified.
- **mpMRI** detects high-grade and anterior tumours (15–33%).
- **PROMIS**: for clinically significant cancer (csPCa), sensitivity was **93%**, specificity 41% and negative predictive value 89%. So MRI lesions need biopsy, and MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume**, so a margin is needed: **5 mm** ablates >95% and 6 mm 98.6% of csPCa. About **90%** of outfield csPCa lies within **1 cm** of the target (the "penumbra").
- Prostate-specific membrane antigen (PSMA) PET is more specific; mpMRI is more sensitive.

> **KEY POINT**
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland — every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI plus targeted and systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal histology
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, tightened from 5 mm)
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually goes to AS

- The goal has moved from eradicating all cancer (2009) to eradicating csPCa (2013–16) to treat-and-retreat to **delay or avoid radical therapy** (2019). FT is investigational, ideally within a trial.

3 · Ablation patterns and energies

The route and energy are chosen by where the lesion sits.

- FT means **lesion, quadrant or hemiablation**. "Hockey-stick" (subtotal) ablation is partial-gland ablation, not FT.
- The transrectal route reaches the posterior gland; transperineal and transurethral routes reach the anterior gland.
- P2 P4 HIFU; Cryotherapy for carcinoma prostate — *HIFU ≥55 °C, posterior lesions; cryo Joule–Thomson, urethral warmer, fistula ≤0.8%*

Not in the paper answer:

- Focal cryotherapy 5-yr biochemical progression-free survival (BPFS) ~62–70%; COLD registry: partial equals whole-gland control with ~2× better erections.
- Focal salvage results after radiotherapy and FORECAST (§7); failure definitions and PI-FAB (§6).

Modality	Mechanism / route	Pros	Cons
TULSA (MRI-guided transurethral ultrasound ablation)	Transurethral transducer with MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin, so **highest infield failure** (~40% at 6 yr)
Irreversible electroporation (IRE)	Non-thermal pulses >1600 V/cm open membrane pores; probes 10–20 mm apart	Spares connective tissue; best function vs matched robotic prostatectomy (RARP)	Needs deep anaesthesia (muscle contraction); NVB and urethral injury on histology
Vascular-targeted photodynamic therapy (VTP)	IV padeliporfin (WST11, TOOKAD) 4 mg/kg activated by 753-nm light occludes vessels	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%

Modality	Mechanism / route	Pros	Cons
Focal brachytherapy	Low-dose-rate seeds or high-dose-rate (HDR) catheters	Mild early toxicity	HDR: stricture ≤8.2%, erectile dysfunction (ED) in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants (cryo-enhancers, stereotactic radiotherapy boost) all aim to make that edge lethal. Hence laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Function is well kept by all energies; infield cancer control varies widely.

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
High-intensity focused ultrasound (HIFU)	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil to mild fall

- Cryotherapy 5-yr BPFS is ~62–70%. In COLD, partial and whole-gland cryotherapy gave equal BPFS, with ~2× better erections after partial.
- HEAT (HIFU): 7-yr failure-free survival was **69%** (68% intermediate, 65% high risk).

5 · Complications

Complications are few, but fistula is the one to fear.

- The commonest is **urinary retention** (0–9%); others are infection, sloughing and stricture (~4–5% after HIFU or IRE).
- **Rectourethral fistula (RUF)** occurs in ≤2% and is the most feared; some heal with a catheter for 4–6 weeks. ED is worst after focal HDR.

6 · Follow-up and failure

Follow-up relies on MRI and biopsy, not PSA.

- Prostate-specific antigen (**PSA**) falls only **50–80%**, and the Phoenix/ASTRO definitions are not validated after FT. A rise to nadir + **1.0** ng/mL at 12 months, or + 1.5 at 24–36 months, predicts csPCa. Digital rectal examination is unreliable.

Success (ICUD–SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

1. At **3–6 months**: mpMRI and targeted biopsy (4–6 cores) of the treated zone.
2. At 1–2 and 5 years: mpMRI; if a new lesion appears, then targeted biopsy.
3. At 1–2 and 5 years: systematic biopsy of the untreated gland.

- **PI-FAB** (Prostate Imaging after Focal Ablation): read dynamic contrast-enhanced (**DCE) images first**, as scar obscures T2 and diffusion (DWI). Score 1 (both negative): monitor. Score 2 (DCE positive only): biopsy if PSA rises. Score 3 (both positive): biopsy.

7 · Salvage

Failure after FT or after radiotherapy can be salvaged.

- After FT failure, options are **focal retreatment** or whole-gland RP or radiotherapy. † Salvage RP is feasible but harder (fibrosis), with worse continence and potency than primary RP.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	Progression-free survival 60%, metastasis-free survival 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	RUF ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%
Trial	FORECAST (cryo vs HIFU, ongoing)		SAFE pilot

- Focal salvage controls cancer as well as whole-gland salvage, with better function.

8 · Landmark trials

Evidence for FT is still mostly registry data.

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy at 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems: can insignificant satellites be safely left; cost of repeated treatment; adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy — prostate FT relies on surveillance and retreatment instead.

164 Treatment of Locally Advanced Prostate Cancer

- P2 P1 Staging and management of locally advanced prostate cancer; Staging and management of non-organ-confined CA prostate — *definition, risk groups, work-up, RT + ADT, RP, salvage RT, pN1, algorithm*

Not in the paper answer:

- Progression risk by pathology after prostatectomy (margin highest, 4.37); neoadjuvant hormone limits; CALGB 90203 (§3).
- pN1: hormones plus RT beat hormones alone (OS 74% vs 55%) (§3).
- RTOG 86-10, Canadian biopsy trial, bicalutamide 150 mg harm (HR 1.23), cryoablation vs RT (§4–5).

1 · Definition and natural history

No single treatment cures most of these men, so therapy is combined.

- The working definition is **T3–4, N0 or N1, M0**, and it calls for **multimodal therapy**.
- National Comprehensive Cancer Network (NCCN) high risk is cT3a, prostate-specific antigen (PSA) >20 or grade group (GG) 4–5. † Very high risk is cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores of GG 4–5.
- Because of stage migration, high risk now mostly means **high grade**, not a palpable T3.
- **Grade drives death**: untreated Gleason 8–10 had 15-yr cancer mortality of 60–87% before PSA; it is still >20%.
- In a Swedish cohort, T3–T4 progressed as often as T1–T2 (~48%), but disease-specific survival (DSS) was 57% vs 81%.

2 · Staging and risk assessment

Combine all tools, because each underestimates extent.

- Clinical staging and imaging both **understage**. Partin tables apply to localised disease only. The Kattan radical prostatectomy (RP) nomogram does not apply to cT3b or node-positive disease.
- In CAPRA, each 2-point rise doubles recurrence risk. Decipher reclassifies ~⅔.
- NCCN orders a bone scan and CT or MRI for cT3–T4, PSA >20, GG 4–5 or primary pattern 5. Prostate-specific membrane antigen (PSMA) PET may substitute.
- **PSMA PET** has nodal specificity ≥95% but sensitivity only ~40–66%. In **proPSMA** its accuracy was **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

Surgery removes the primary and stages exactly, if all tumour can be excised.

- **Clinical cT3 is overstaged in 7–27%**, and >⅓ of high-risk specimens are confined. So do not exclude men on clinical stage alone.
- NCCN advises RP with extended pelvic node dissection (**ePLND**) only if the tumour is **completely excisable**; otherwise androgen deprivation therapy (ADT; "AD" in trial reports) with radiotherapy (RT).
- In cT3, about half are recurrence-free at 8–10 yr on surgery alone. DSS is **~80–90% at 10 yr**, though biochemical disease-free survival is often ~50%.

KEY POINT

Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37) — higher than extracapsular extension (ECE), seminal vesicle invasion (SVI) or nodes — so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

- pT3bN0 at 10 yr: biochemical recurrence (BCR)-free ~25%, metastasis-free survival (MFS) 70%, DSS 80%.
- In **node-positive** disease with a single node, nearly half never progress.

Neoadjuvant AD before RP

- PSA falls ~90% and volume 30–50%. **Clinical downstaging occurs in 32–90%, but pathological in only 8–31%**.
- It reduces margins in cT1–T2 but not cT3. It has **no** effect on SVI, nodes, PSA recurrence, DSS or overall survival (OS).
- 8 vs 3 mo lowered margins from 23% to 12%, with no PSA benefit.
- **CALGB 90203** (AD plus docetaxel before RP) missed its primary endpoint but improved MFS and OS. Complete response (pT0) is 0–4%.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or positive margin; immediate RT vs observation	Biochemical disease-free survival (bDFS) 74% vs 53%; **benefit only with positive margins**
SWOG 8794	Same	**MFS (hazard ratio [HR] 0.71) and OS (HR 0.72)** — the only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% in both arms

- **Trap**: very high-risk men (Gleason 8–10, pT3b) were only ~20% of the timing trials. So adjuvant RT may still help them, and extensive or multiple margins also justify it.
- Dose is **≥64 Gy**. **GETUG-AFU 16**: **3 months** of AD with salvage RT gave 10-yr progression-free survival (PFS) 64% vs 49% (HR 0.54) *(see Ch 162)*.

Adjuvant AD and pN1

- In **pT3N0**, adjuvant AD reduces progression but gives **no DSS or OS gain**.
- **ECOG 7887** (nodes found at RP): immediate AD improved OS, DSS and PFS.
- **AD plus RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%; OS HR ~0.4–0.46. AD alone raises other-cause mortality (HR 3.05).
- Adjuvant docetaxel or mitoxantrone after RP gives no benefit.

ALGORITHM

pN1 after RP

1. If favourable (few nodes, Gleason ≤8), then observe with PSA monitoring.
2. If adverse (>3 nodes, Gleason >8), then give AD plus RT (best OS).
3. Avoid AD alone in a fit man.

4 · RT + ADT

This is the standard for most men.

- The standard is **dose-escalated RT (>72, ideally ~78–81 Gy) with ADT**, often with pelvic RT *(see Ch 162)*.
- With RT, ADT shrinks the target and is synergistic. Canadian trial: positive 24-mo biopsy 65% (RT alone), 28% (plus 3 mo neoadjuvant AD [NAD]), 5% (neoadjuvant, concurrent and adjuvant AD).

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality fell from 36% to 23%; **OS unchanged**
TROG 96.01	RT with no, 3 or 6 mo NAD	6 mo lowered all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control and MFS; OS gain only in Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better quality of life
RTOG 94-13	Pelvic vs prostate RT, and neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis with adjuvant AD had worst OS

- **Duration**: intermediate risk 4–6 mo; high risk (NCCN) **18–36 mo** with external beam RT (EBRT), 12–36 mo with a brachytherapy boost.
- **ASCENDE-RT**: a low-dose-rate (LDR) boost halved biochemical recurrence vs an EBRT boost (HR 2.04; 9-yr 85% vs 65%), with equal OS. In Gleason 9–10, EBRT with a boost lowered cancer mortality (HR ~0.4).
- **Trap**: AD plus RT benefits often exclude OS. The OS gain is seen with long AD in high risk (EORTC 22863: local control 97% vs 79%; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death relative risk (RR) **0.44**; overall mortality RR 0.68.
- **PR.3/PR07**: DSS HR 0.54; 7-yr OS rose from 66% to 74%.
- So RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD and RT	OS HR 0.69
GETUG 12	Docetaxel and estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel, estramustine and etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone and prednisolone for 2 yr**, added to 3 yr AD and RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4 and PSA ≥40**. **NCCN 2023 advises RT, ADT and 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

ADT alone is for men who cannot or will not have local treatment.

- **MRC**: immediate rather than deferred AD gave fewer cancer deaths, mainly in M0. Immediate vs deferred orchiectomy (in men declining local therapy) gave equal OS.
- **Bicalutamide 150 mg** improved PFS in locally advanced disease but **raised death in localised disease** (HR 1.23), so never use it at low risk.
- AD alone does not stop local progression: over half need a transurethral resection (TURP), and ureteric obstruction and haematuria follow.
- **Cryoablation** vs RT (randomised): equal DSS and OS, but worse biochemical control (failure 87% vs 53%).
- **RP vs RT with ADT**: there is no completed randomised trial; **SPCG-15** (cT3N0M0) is ongoing.
- After RT alone for cT3: 10-yr local recurrence 24–74%, TURP ~⅓, hydronephrosis 20%.

ALGORITHM

Locally advanced / high-risk M0

1. Stage with PSMA PET, or CT and bone scan.
2. If fit and resectable, then RP with ePLND. If pathology is adverse, then early salvage RT (adjuvant if pT3b, Gleason 8–10 or positive margins). If pN1, then observe or give AD plus RT.
3. Alternative: dose-escalated RT (with or without brachytherapy boost and pelvic nodes) plus 18–36 mo ADT.
4. If very high risk or N1, then add 2 yr abiraterone.
5. If unfit for local therapy, then ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" — SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

The trials to quote in answers.

Trial	Design	Result	Take-home
SPCG-7	ADT with or without RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT with or without RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT with or without 3 yr goserelin	5-yr OS 62% vs 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain in Gleason 8–10	Prolong ADT for high grade

Trial	Design	Result	Take-home
STAMPEDE (abiraterone)	ADT and RT, with or without 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS and OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional imaging	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

Act early enough to salvage, not so early as to overtreat.

Biochemical recurrence (BCR) is a rise in prostate-specific antigen (PSA) after curative therapy, before clinical recurrence. Not every detectable PSA progresses, but clinical recurrence is almost always preceded by one.

- After radical prostatectomy (RP), **¼–⅖** develop BCR within 10 years. Only ≈half of intermediate/high-risk men with BCR fail clinically.
- At Memorial Sloan Kettering (MSK), PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis and secondary therapy.
- **Ultrasensitive PSA** is prognostic, not diagnostic. About half of early detectable values fall on retesting, so repeat the test rather than treat.

Setting	Definition	Logic
Post-RP — American Urological Association (AUA)	PSA **≥0.2 ng/mL**	No gland remains, so any PSA is suspect
Post-RP — National Comprehensive Cancer Network (NCCN)	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1	Any of three triggers
Post-RP — European Association of Urology (EAU)	No threshold; risk groups (below)	Kinetics plus grade
After external beam RT (EBRT), with or without androgen deprivation therapy (ADT)	**Phoenix**: **nadir + 2 ng/mL**	The remnant gland makes PSA; validated against metastasis and death
After low-dose-rate (LDR) brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	≥2× the EBRT dose gives a very low nadir, so Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After radiotherapy (RT)
Low risk (needs BOTH)	PSA doubling time (PSADT) **>1 year** AND grade group (GG) <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

KEY POINT

PSA bounce after RT: a transient rise of 0.1–0.5 ng/mL at a mean of ≈ **9 months** (radiation prostatitis, delayed cell death), followed by a fall. It is **favourable** (better BCR-free survival), so monitor; never salvage on a bounce.

2 · Risk prediction and natural history

Most men with BCR never die of it, so risk decides treatment.

- A **positive margin** is the only modifiable BCR predictor (pT3a margin-negative did better than pT2 margin-positive). Genomic classifiers are unproven for selecting salvage.
- **CAPRA** is the pre-operative score and **CAPRA-S** the post-operative one MCQ. Only the MSK nomograms predict 15-year cancer-specific survival.
- **Pound**: untreated, a median **8 years** passes from BCR to metastasis, then **5 years** to death (P2-144).
- After Phoenix failure, metastasis follows at ≈5.4 years and cancer death at ≈10.5. Follow-up is lifelong, because ¼ recur after 5 years.

3 · Imaging at recurrence

*Purpose: **exclude metastasis before further local therapy**, then localise local recurrence.*

- P2 65 yr male underwent RP… PSA has risen — *BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis*

Not in the paper answer:

- MSK threshold, repeat ultrasensitive PSA, LDR cure <0.2 at 4 years (§1); CAPRA-S, ¼ recur after 5 years (§2).
- EMPIRE-1; PSMA PET changes the field in ≈1 in 5; negative mpMRI predicts salvage RT failure (§3–4).
- Selection for salvage after RT (§5); intermittent ADT unproven when metastatic (§6).
- Bone scan and CT are rarely positive below PSA **10 ng/mL**.
- **Prostate-specific membrane antigen (PSMA) PET** is positive in ≈½ at PSA **<0.5**, and finds disease outside the salvage field in ≈1 in 5. In **EMPIRE-1**, fluciclovine-guided salvage improved 3-year control.
- Multiparametric MRI (mpMRI) is best for local recurrence. After RT, dynamic contrast (DCE) and diffusion (DWI) sequences carry the diagnosis. **Biopsy after RT** is mandatory before local salvage.

> **KEY POINT**
> Salvage RT works best at the lowest PSA, but imaging detection and certainty of recurrence improve only as PSA rises. Risk features (grade, margins, kinetics) let us act early without overtreating.

4 · Salvage after radical prostatectomy

Bed RT is the only cure here.

- **Salvage RT (SRT)** is the only curative option (P2-144). Give **≥64–65 Gy** to the fossa, using intensity-modulated RT (IMRT).
- Delay costs control: ≈10% relapse-free survival is lost per 0.1 ng/mL in men with ≥2 adverse features.
- No visible lesion on mpMRI predicts SRT failure, because the disease is probably occult metastasis.

5 · Salvage after radiotherapy

Irradiated tissue heals badly, so select carefully.

Prerequisites: biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years and a shared decision. Every modality gives

≈**50–70%** control, so selection (local-only disease, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2. Gives pathological staging; poor if seminal vesicles or nodes are involved	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ high-dose-rate)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage stereotactic body RT (SBRT)	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; a post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), erectile dysfunction; incontinence single-digit with **urethral warming**
Salvage high-intensity focused ultrasound (HIFU)	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy** is worked up like RT failure: PET, then MRI, then targeted biopsy, then any salvage option.

6 · Non-curative approach to BCR: observation vs ADT

ADT delays metastasis but costs toxicity and drives resistance.

- AUA: observation (or a trial) after maximal local therapy; ADT only for higher metastatic risk (PSADT **<12 months** or high grade), preferably intermittent.
- EAU advises observing low-risk BCR, limited life expectancy, or men declining local therapy.
- Primary ADT for localised disease is **not** recommended.
- **Messing (ECOG)**: immediate ADT for node-positive (pN+) disease after RP improved overall survival (OS) and progression-free survival (PFS). But it was small, pre-dated modern imaging, had bulky nodes, and treated the delayed arm only at clinical progression.
- Low-volume pN+ with undetectable PSA: **20–40%** are cured by surgery alone.
- MRC: immediate ADT reduced progression-related symptoms, but the OS gain faded. TOAD (rising PSA): OS hazard ratio (HR) 0.55. EORTC 30891: benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, with PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent was **non-inferior** (OS 8.8 vs 9.1 years) with better quality of life.
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23). This was **inconclusive**, as the interval exceeded the 1.20 margin, so intermittent may be worse in metastatic disease.

7 · Androgen deprivation therapy: biology and agents

Know each agent's action and side effect.

Castration removes the ligand, not the receptor. So castration-resistant prostate cancer (CRPC) keeps androgen receptor (AR) signalling alive through AR amplification (20–30%), ligand-binding-domain mutations (which turn an antagonist into an agonist), splice variants (**AR-V7**, poor prognosis), steroid synthesis within the tumour, and glucocorticoid receptor bypass.

Castrate testosterone: <50 ng/dL.

- P4 P2 GnRH (LHRH) analogues; Maximum androgen blockade and bicalutamide — *agonist flare/FSH rise, antagonists (degarelix, relugolix/HERO), antiandrogen toxicities, CAB, withdrawal response*

Not in the paper answer:

- Resistance mechanisms (AR-V7); orchiectomy safety advantage; HERO day-4 castration; ketoconazole; ARTA withdrawal 0–10% (table and pearl below).
- FRAX drug thresholds and anaemia (§8).

Method	Key facts
Bilateral orchiectomy	≈95% of testosterone is testicular; castrate within hours (median ≈15 ng/dL). Fewer cardiac events, fractures and peripheral arterial disease than agonists, with equal survival, so offer it for lifelong ADT
Luteinising hormone-releasing hormone (LHRH) agonists / antagonists	See P4-103. HERO: castrate by day 4 in 56% vs 0%
Cyproterone acetate	Inferior to goserelin; cardiovascular events up to 10%, so not used for cancer; 50–100 mg/day for hot flushes
AR-targeting agents (ARTAs): enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures with enzalutamide; rash and hypothyroidism with apalutamide; darolutamide has the fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Abiraterone lowers cortisol, which raises ACTH and causes mineralocorticoid excess (low potassium, hypertension, oedema), so give it with prednisone. Ketoconazole castrates within 4 h: an emergency option

KEY POINT

Flare matters with spinal or weight-bearing bone metastases: give an antiandrogen with (or before) the agonist for **21–28 days**; a simultaneous start is as good as a week's lead-in. For **impending cord compression** or severe pain, use an **antagonist or orchiectomy** (or ketoconazole).

Antiandrogen withdrawal: 15–30% respond (bicalutamide); 0–10% with ARTAs; no OS gain.

8 · ADT side effects and their management

Protect bone, metabolism and heart from the start.

Effect	Key facts	Management
Osteoporosis / fracture	Caused by loss of peripheral oestrogen; fracture risk rises 20–45%	Baseline DEXA and FRAX. For all: calcium 1000–1200 mg and vitamin D 400–1000 IU/day, exercise, stop smoking. Give a drug if T-score is −1 to −2.5 with 10-year hip fracture risk **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly. Repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens most effective; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine is ineffective
Metabolic	Fat rises with sarcopenia; atypical metabolic syndrome (subcutaneous fat, raised HDL); diabetes (agonist HR 1.44)	Exercise; monitor glucose and lipids
Cardiovascular	Agonists increase coronary disease, myocardial infarction, sudden death and venous thromboembolism	Risk-factor control; orchiectomy or antagonist favoured
Anaemia	Normocytic; most on combined blockade (CAB) lose ≥10% Hb within 6 months; pre-ADT Hb <12 g/dL predicts shorter survival	Check blood count; reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy or tamoxifen; established cases need surgery

Follow-up on ADT

Check PSA, blood count and chemistry 3–6-monthly. **If PSA rises, check testosterone first.** Inadequate suppression is a drug problem (switch agent or do orchiectomy). A rising PSA with castrate testosterone means **CRPC** *(see Ch 166)*, so restage. The germline *HSD3B1* (1245A>C) variant predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

ADT alone is not enough; volume and fitness decide what is added.

- P3 Management of newly Metastatic prostate cancer — *CHAARTED/LATITUDE definitions, doublets and triplets with HRs, prostate RT for low volume, MDT, genetic testing, follow-up*

Not in the paper answer:

- Prognosis by timing, volume and 7-month PSA nadir (below).
- Genes to test: germline BRCA1/2, ATM, PALB2, CHEK2, HOXB13, mismatch repair; somatic CDK12, MSI-H.
- Metachronous low-volume disease has the best prognosis (median OS ≈92 months vs ≈43 for synchronous high volume). Nodes do better than bone, and bone better than visceral.
- **SWOG 9346 PSA nadir at 7 months**: ≤0.2 gives median OS **75** months; 0.2–4 gives 44; >4 gives 13.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral metastasis (no grade criterion)	Docetaxel benefit; RT to primary in low volume
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term CAB is not recommended.
- Docetaxel is **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77).
- Observe only if asymptomatic with life expectancy <5 years.
- The benefit of prostate RT on top of modern combinations is unknown. Cytoreductive prostatectomy is for trials or palliation only.
- **Metastasis-directed therapy (MDT)**: oligometastases are **≤3 sites** on PET, treated by SBRT or salvage node dissection.

10 · Landmark trials

These trials set salvage and metastatic practice.

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed plus 4–6 months ADT vs pelvis plus ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT and pelvic nodes (not OS)
RTOG 9601	SRT with or without 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT with or without 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); metastasis-free survival (MFS) HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide with or without enzalutamide vs enzalutamide	MFS 87.3% (combination) / 80.0% (enzalutamide) / 71.4% (leuprolide); combination HR 0.42, enzalutamide alone 0.63	Enzalutamide for high-risk non-metastatic HSPC
CHAARTED (ECOG 3805)	ADT with or without docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT with or without docetaxel (up to 9 cycles)	OS HR 0.88, not significant; better PFS	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic and high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, 3-year OS 83 vs 76%, failure-free survival HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT with or without abiraterone	OS HR 0.62; radiographic PFS (rPFS) 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT with or without enzalutamide	rPFS HR 0.39	Volume-independent

Trial	Design	Result	Take-home
ENZAMET	ADT plus enzalutamide vs ADT plus older antiandrogen	OS HR 0.67	Control was not placebo
TITAN	ADT with or without apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2, with or without abiraterone and prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT and docetaxel, with or without darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT with or without prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

Why this matters: the androgen receptor still drives growth, which explains treatment order.

- P2 Castration-resistant prostate cancer (CRPC): definition and treatment — *EAU definition, AR mechanisms, evaluation, general principles, nmCRPC, mCRPC options table, sequencing, supportive care*

Not in the paper answer:

- A prostate-specific antigen (PSA) doubling time (PSADT) **<3 months** predicts a rapid course, so aggressive therapy should be considered.
- Radiographic progression without a PSA rise is common with early ARPI use (ARCHES: up to ⅔), so image periodically.
- Exogenous testosterone can cause a severe flare.
- Symptomatic bone pain or visceral-predominant disease calls for **upfront chemotherapy**; limited, asymptomatic disease suits sequential hormonal therapy.

CRPC means progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. With androgen deprivation therapy (ADT) alone, metastatic disease historically had progression-free survival (PFS) of 12–20 months and overall survival (OS) of 24–36 months.

- **Castration resistance is not androgen independence.** The androgen receptor (AR) still drives growth through AR amplification, enhancer amplification, ligand-binding-domain mutations, the splice variant **AR-V7**, adrenal androgens (up to 10%) and synthesis within the tumour. Hence AR-pathway inhibitors (ARPIs) work.
- Once AR dependence is lost, use taxanes, radium-223, prostate-specific membrane antigen (PSMA) therapy, poly(ADP-ribose) polymerase inhibitors (PARPIs) and immunity.

2 · Clinical states and progression criteria

Why this matters: early flare mimics progression, so firm rules prevent stopping a working drug.

Tumour–node–metastasis (TNM) staging does not guide advanced disease. The **Prostate Cancer Clinical Trials Working Group 3 (PCWG3)** clinical states model (2016) classifies men by response to prior therapy: localised disease, rising PSA, non-metastatic CRPC (nmCRPC), then metastatic CRPC (mCRPC) in first, second and later lines.

- † PCWG3 PSA progression is a rise of ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.

Modality	Pitfall	PCWG3 rule
PSA	Chemotherapy may cause a rise before a fall; little change with sipuleucel-T or radium-223; flare with radium-223 and ^{177}Lu-PSMA	Confirm any rise
Bone scan	Early new lesions may be **flare** of healing disease	Rescan at **12–16 weeks** as new baseline; progression = **≥2 new lesions** ("2 + 2")
Computed tomography (CT)	Soft tissue is a small fraction of burden	RECIST criteria; PFS preferred over response rate

Modality	Pitfall	PCWG3 rule
PSMA positron emission tomography (PET)	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

KEY POINT
Every marker (PSA, bone scan, PSMA PET) can flare early, so progression is called on confirmed, combined evidence — otherwise effective therapy is abandoned too soon.

- **Continue ADT indefinitely** during every later therapy.

3 · Germline and somatic testing

Why this matters: a mutation can open a targeted drug and affects the family.

- **Germline testing** suits all metastatic disease, a family history of *BRCA1/2*-, Lynch- or *HOXB13*-associated cancers, and Ashkenazi ancestry.
- **Somatic** sequencing uses tissue, usually the archival primary. Liquid biopsy is an alternative, but **clonal haematopoiesis (CHIP)** can give false DNA-repair calls, and such men do not respond to PARPIs.
- Somatic homologous recombination repair (HRR) mutations occur in **20–25%** of mCRPC. Mismatch repair deficiency or microsatellite instability (dMMR/MSI-H) occurs in **2–5%** and should prompt Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

Why this matters: treating a fast-rising PSA early delays metastasis and prolongs life.

nmCRPC is a rising PSA on castration with no metastases on **conventional** imaging. PSMA PET is shrinking this group by finding metastases. PSA kinetics are the strongest predictor; a baseline PSA >10 also predicts. Median time to bone metastasis is ≈30 months.

Add an ARPI (apalutamide, enzalutamide or darolutamide) to ADT when PSADT is ≤10 months. All three improve metastasis-free survival (MFS) and OS. Abiraterone is active (IMAAGEN) but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Why this matters: these oral drugs usually come first; toxicity guides choice.

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase and 17,20-lyase)	1000 mg **fasting** with prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct potassium first. Food increases absorption up to 10× (250 mg with a low-fat meal is an alternative)
Enzalutamide	Pure AR antagonist; blocks translocation and DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	Does not cross the blood–brain barrier (BBB)	240 mg	**Rash, hypothyroidism**, fractures

Agent	Mechanism	Dose	Toxicity / notes
Darolutamide	Structurally distinct; does not cross the BBB	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: blocking CYP17 lowers cortisol, so adrenocorticotropic hormone (ACTH) rises and causes mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid (prednisone may activate a promiscuous AR). All ARPIs cause fatigue, falls and fractures, so take care in men >75.

6 · Cytotoxic chemotherapy

Why this matters: taxanes are the only cytotoxics that extend life.

Docetaxel binds β-tubulin, stabilising microtubules and causing G2/M arrest; it also impairs AR nuclear transport. **Cabazitaxel** has poor P-glycoprotein affinity, so it works after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** with prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly with prednisone and G-CSF; after docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² with prednisone; palliation	No survival benefit
Platinum (with etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

Estramustine was abandoned. Adding anti-angiogenic or bone drugs to docetaxel never improved OS.

7 · Radiopharmaceuticals

Why this matters: two isotopes prolong life in different disease patterns.

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, so marrow is spared); calcium mimetic	Beta; needs PSMA PET-positive disease
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI and taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89 and samarium-153 only palliate pain.

8 · PARP inhibitors

Why this matters: they work only with a DNA-repair mutation, so test first.

- P4 PARP inhibitors in Prostate cancer — *synthetic lethality, PARP trapping, AR–PARP synergy, BRCA2 vs ATM, approvals, toxicity, key trials*

Not in the paper answer:

- PARP1 also promotes AR signalling, which adds to the ARPI–PARPI synergy.
- Haemoglobin can fall suddenly, so watch blood counts.
- PARPIs are ineffective in unselected men. ATM-deficient tumours may suit ATR inhibitors.

9 · Immunotherapy

Why this matters: prostate cancer is immunologically "cold", so few benefit.

- **Sipuleucel-T** is an autologous dendritic-cell vaccine loaded with a prostatic acid phosphatase–GM-CSF fusion protein. It was the first approved therapeutic cancer vaccine.
- It suits **asymptomatic/minimally symptomatic mCRPC with no visceral metastases or opioid pain**. It prolongs OS **without** PSA or PFS change. Toxicity: infusion fever, rigors.
- Checkpoint inhibitors fail in unselected men. **Pembrolizumab** 200 mg IV 3-weekly is approved tumour-agnostically for **MSI-H/dMMR** or tumour mutational burden **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Why this matters: fast progression with a low PSA needs platinum, not hormonal drugs.

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone with or without nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia (PTH-related protein), brain
Molecular	AR alterations	≥2 of *PTEN* loss, *TP53* mutation, *RB1* loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) with or without radiotherapy (RT)
Survival	Years	**<12 months**

Biopsy a metastasis if rapid progression with low PSA; chromogranin A is not needed.

11 · Sequencing and cross-resistance

Why this matters: most men arrive pre-treated, so the next drug must act differently.

A second ARPI helps only **15–30%**, because of cross-resistance, so **switch mechanism** (CARD). See the P2-140 sequencing table; add pembrolizumab for MSI-H/dMMR and a PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19 (check anticoagulants, antiepileptics).
- † AR-V7 in circulating tumour cells predicts resistance to abiraterone and enzalutamide, not to taxanes.

12 · Complications and palliative care

Why this matters: early action preserves walking, continence and kidneys.

- P2 Metastatic prostate cancer with bone metastases — *cord compression pathway, Patchell, pain RT, obstruction, zoledronic acid vs denosumab, ONJ prevention*

Not in the paper answer:

- Compression may arise from paravertebral soft tissue, despite a normal bone scan.
- Circumoral numbness suggests a skull-base metastasis, which should be irradiated.
- Obstructive uropathy must be **relieved before chemotherapy**, but not every unilateral case needs drainage.
- Erythropoiesis-stimulating agents are not recommended for anaemia.

Every focal back pain in a man with bone metastases is cord compression until proven otherwise.

> **ALGORITHM**
> 1. Back pain with or without deficit or sphincter change: suspect compression.
> 2. Give **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day).
> 3. Arrange **urgent whole-spine MRI**.
> 4. Give RT with or without decompressive surgery.
> 5. Taper steroids over 2–3 weeks.

- Operate if progression during RT, unstable fracture, or recurrence after RT.
- Neuropathic pain needs tricyclics, gabapentinoids or nerve block. Start hospice care in time.

13 · Bone-targeted therapy

Why this matters: they prevent fractures and pain but do not prolong life.

Two uses: (1) ADT osteoporosis; (2) **prevention of skeletal-related events (SREs)** in CRPC bone metastases. There is **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, osteonecrosis of the jaw (ONJ)	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

> **KEY POINT**
> **ONJ** occurs mostly with dental extraction or poor teeth, so check the teeth first. Give calcium 1000 mg and vitamin D 800 IU daily with both agents. Bone agents are strongly recommended with radium-223.

14 · Landmark trials

Why this matters: each approval rests on these trials.

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug
PROSPER	nmCRPC; enzalutamide	MFS improved; death reduced 27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic

Trial	Design	Result	Take-home
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m^2; docetaxel first
CARD	After docetaxel and 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	Radiographic PFS (rPFS) 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death reduced 29%; radiographic progression reduced 81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI and taxane, PSMA-positive; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	*BRCA/ATM* rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	*BRCA* rPFS HR 0.50; no *ATM* benefit	*BRCA* only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI with abiraterone / with enzalutamide	rPFS gains, largest in *BRCA*; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/*BRCA*
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

Why this matters: MRI first avoids needless biopsies.

ALGORITHM

1. **Raised prostate-specific antigen (PSA) or abnormal digital rectal examination (DRE)**: repeat it; exclude infection, instrumentation, 5α-reductase inhibitors (double PSA, ×2); use density, biomarkers (→ *Ch 154*).
2. **Multiparametric MRI (mpMRI) before biopsy**: PI-RADS (→ *Ch 155*).
3. **Biopsy** PI-RADS 3–5 targets plus systematic cores; transperineal is safer.
4. Assign **ISUP Grade Group (GG)** (→ *Ch 156*) and risk group (→ *Ch 157*). If unfavourable intermediate risk or worse, then stage with PSMA PET (or CT and bone scan).

B · Localised disease: matching risk to treatment

Why this matters: risk decides watching versus treating.

National Comprehensive Cancer Network (NCCN): favourable groups need ALL criteria, adverse groups ANY one. Weigh life expectancy, comorbidity, preference.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	Active surveillance (AS)
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	Radical prostatectomy (RP) or radiotherapy (RT); AS (low-volume GG2); focal in trials
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT with 4–6 months androgen deprivation therapy (ADT)
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP with extended pelvic node dissection (ePLND), or RT with ADT 18–36 months; add abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT and ProtecT show that radical treatment helps only younger men with ≥intermediate-risk disease and long life expectancy. ProtecT: RP = RT for 15-year cancer survival.

C · Surgery vs radiotherapy vs focal therapy

Why this matters: cure rates are similar, so side effects decide.

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, erectile dysfunction (ED)	Rectal/urinary irritation; late ED; ADT	Least; retreatment common
Advantages	Pathological staging; salvage RT	No anaesthesia; comorbid men; hypofractionation	Preserves function; MRI-visible index lesion
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Salvage	Bed RT, early at PSA 0.1–0.2, with or without ADT	RP, cryotherapy, HIFU, brachytherapy	Repeat focal, whole-gland, RP, RT

D · Advanced disease

Why this matters: intensify early; switch mechanism at resistance.

ALGORITHM

1. **Biochemical recurrence (BCR)** (Ch 165): PSMA PET. After RP: salvage RT. After RT: local salvage if biopsy-proven. If high risk (PSA doubling time ≤9 months), then give ADT and enzalutamide (EMBARK).
2. **Metastatic hormone-sensitive disease (mHSPC)**: never ADT alone; add an androgen-receptor pathway inhibitor (ARPI), or a docetaxel triplet if fit and high volume. If low volume, add prostate RT (STAMPEDE arm H). Offer genetic testing to all.
3. **Non-metastatic castration-resistant disease (nmCRPC)** with PSA doubling time ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).
4. **Metastatic castration-resistant disease (mCRPC)** (Ch 166): switch mechanism. Docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 for symptomatic bone-only disease (ALSYMPCA); a PARP inhibitor if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H. Always protect bone.